Optimization Methods Colloquium

Answer key for tickets 1–20 (examiner only)

Ticket 1

Definitions

1. **Strict local minimum.** Like a local minimum, but with strict inequality: $f(x^\star) < f(x)$ for all $x \in \mathcal{X} \cap B_r(x^\star)$, $x \neq x^\star$.
2. **Positive definite and positive semidefinite matrix.** Symmetric H: $H \succeq 0$ if $x^\top H x \geq 0$ for all x; $H \succ 0$ if $x^\top H x > 0$ for all $x \neq 0$. Equivalently, all eigenvalues are ≥ 0 (respectively > 0). For 2×2, Sylvester's criterion: $H \succ 0 \iff h_{11} > 0,\ \det H > 0$.
3. **L-smoothness (Lipschitz gradient).** $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$ for all x, y. For $f \in C^2$ it suffices that $\|\nabla^2 f(x)\|_2 \leq L$. Consequence (descent lemma): $f(y) \leq f(x) + \langle \nabla f(x), y - x\rangle + \frac{L}{2}\|y - x\|^2$. For a quadratic, $L = \lambda_{\max}(Q)$.
4. **Lagrangian.** For $\min f(x)$ subject to $g_i(x) \leq 0$, $h_j(x) = 0$: $\mathcal{L}(x, \lambda, \nu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \nu_j h_j(x)$, where $\lambda_i \geq 0$ and ν_j are unrestricted in sign.
5. **Karush–Kuhn–Tucker (KKT) conditions.** Stationarity $\nabla_x \mathcal{L}(x^\star, \lambda^\star, \nu^\star) = 0$; primal feasibility $g_i(x^\star) \leq 0$, $h_j(x^\star) = 0$; dual feasibility $\lambda_i^\star \geq 0$; complementary slackness $\lambda_i^\star g_i(x^\star) = 0$. For convex f, g_i and affine h_j they are sufficient for global optimality.
6. **Dual function and dual problem.** $q(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu)$ is always concave (an infimum of affine functions). Dual problem: $\max_{\lambda \geq 0, \nu} q(\lambda, \nu)$. Weak duality: $d^\star \leq p^\star$.

Theoretical results

1. The negative gradient is the direction of steepest local decrease (among $\|d\|_2 = 1$).
Proof. The rate of change along a unit d is $D_d f(x) = \langle \nabla f(x), d\rangle$. By Cauchy–Schwarz, $\langle \nabla f(x), d\rangle \geq -\|\nabla f(x)\|\|d\| = -\|\nabla f(x)\|$, with equality for $d = -\nabla f(x)/\|\nabla f(x)\|$. So this direction gives the smallest (most negative) derivative. □

2. Guaranteed decrease of a GD step for L-smooth f: $f(x^+) \leq f(x) - \alpha(1 - \frac{L\alpha}{2})\|\nabla f(x)\|^2$.
Proof. Descent lemma (may be used without proof): $f(y) \leq f(x) + \langle \nabla f(x), y - x\rangle + \frac{L}{2}\|y - x\|^2$. Substitute $y = x^+ = x - \alpha \nabla f(x)$: $f(x^+) \leq f(x) - \alpha\|\nabla f(x)\|^2 + \frac{L\alpha^2}{2}\|\nabla f(x)\|^2 = f(x) - \alpha\left(1 - \frac{L\alpha}{2}\right)\|\nabla f(x)\|^2$. Hence for $0 < \alpha < 2/L$ there is strict decrease away from stationary points, and for $\alpha = 1/L$: $f(x^+) \leq f(x) - \frac{1}{2L}\|\nabla f(x)\|^2$. □

Problem 1: type of a stationary point

Find the stationary points of $f(x, y) = 3x^2 - 2xy + 3y^2 - 4x - 4y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x} = 6x - 2y - 4 = 0, \quad \dfrac{\partial f}{\partial y} = -2x + 6y - 4 = 0.$ The system is linear with determinant $32 \neq 0$, so the unique solution is $x^\star = (1,\ 1)$.

2. Hessian. $H = \begin{pmatrix} 6 & -2 \\ -2 & 6 \end{pmatrix}$ (constant). Sylvester's criterion: $h_{11} = 6 > 0$, $\det H = 6 \cdot 6 - 2^2 = 32 > 0$, hence $H \succ 0$.

Via eigenvalues: $\lambda^2 - 12\lambda + 32 = 0$, $\lambda_1 = 4,\ \lambda_2 = 8$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H (x - x^\star) > 0$ for $x \neq x^\star$, so the minimum is global and unique. $f(x^\star) = -4$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 4$, $L = \lambda_{\max}(H) = 8$, $\kappa = L/\mu = 2$. f is 4-strongly convex and 8-smooth. GD converges for $0 < \alpha < 2/L = \frac{1}{4}$; the standard step is $1/L = \frac{1}{8}$, the optimal step for a quadratic is $2/(L+\mu) = \frac{1}{6}$ with factor $(\kappa-1)/(\kappa+1) = \frac{1}{3}$.

Problem 2: gradient descent

For $f(x,y) = x^2 - xy + y^2$, perform three steps of gradient descent from $x_0 = (6,\ 0)$ with step $\alpha = \frac{1}{3}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (2x - y,\ -x + 2y)$, $H = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, eigenvalues $\mu = 1$, $L = 3$.

Minimizer $x^\star = (0,\ 0)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{3} < 2/L = \frac{2}{3}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{3}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(6,\ 0)$	36	$(12,\ -6)$
1	$(2,\ 2)$	4	$(2,\ 2)$
2	$\left(\frac{4}{3},\ \frac{4}{3}\right)$	$\frac{16}{9}$	$\left(\frac{4}{3},\ \frac{4}{3}\right)$
3	$\left(\frac{8}{9},\ \frac{8}{9}\right)$	$\frac{64}{81}$	—

Why it behaves this way. Eigendirections of H: $(1,1)$ with $\lambda = 1$ and $(1,-1)$ with $\lambda = 3$. Each step multiplies the error components along them by $1 - \frac{1}{3}\cdot 1 = \frac{2}{3}$ and $1 - \frac{1}{3}\cdot 3 = 0$ respectively. A factor of 0 means this component is eliminated in a single step (step $\alpha = 1/\lambda$). The overall speed is set by the largest $|1 - \alpha\lambda| = \frac{2}{3}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{2}{3}\right)^2 = \frac{4}{9}$.

Ticket 2

Definitions

1. **Optimization problem: decision variable, objective, feasible set.** A problem $\min_{x\in\mathcal{X}} f(x)$, where x is the decision variable (e.g. model weights), f is the objective (error, cost, risk), and $\mathcal{X}$ is the feasible set (constraints; without constraints $\mathcal{X}=\mathbb{R}^d$). Maximizing f is equivalent to minimizing $-f$.
2. **Local minimum.** $x^\star\in\mathcal{X}$ is a local minimizer if there exists $r>0$ such that $f(x^\star)\le f(x)$ for all $x\in\mathcal{X}\cap B_r(x^\star)$.
3. **Global minimum.** $x^\star\in\mathcal{X}$ is a global minimizer if $f(x^\star)\le f(x)$ for all $x\in\mathcal{X}$. A minimum may fail to exist (e^x on $\mathbb{R}$) and may be non-unique ($(x^2-1)^2$).
4. **Gradient (via the linear approximation).** For differentiable $f:\mathbb{R}^d\to\mathbb{R}$, the gradient $\nabla f(x)$ is the vector giving the linear part of the increment: $f(x+h)=f(x)+\langle\nabla f(x),h\rangle+o(\|h\|)$. Equivalently, $df=\langle\nabla f(x),dx\rangle$.
5. **Stationary point and saddle point.** A stationary point satisfies $\nabla f(x)=0$. A saddle point is a stationary point that is neither a local minimum nor a local maximum (there are directions of both increase and decrease; e.g. an indefinite Hessian, as for x^2-y^2 at the origin).
6. **Convex function.** f on a convex set C is convex if $f((1-t)x+ty)\le(1-t)f(x)+tf(y)$ for all $x,y\in C$, $t\in[0,1]$. For differentiable f this is equivalent to $f(y)\ge f(x)+\langle\nabla f(x),y-x\rangle$, and for C^2 functions to $\nabla^2 f\succeq 0$.

Theoretical results

1. First-order necessary condition: at an unconstrained local minimizer $\nabla f(x^\star)=0$.

Proof. Fix a direction d and let $\phi(t)=f(x^\star+td)$. Since $x^\star$ is a local minimizer of f, $t=0$ is a local minimizer of the one-variable function ϕ, so $\phi'(0)=0$. By the chain rule, $\phi'(0)=\langle\nabla f(x^\star),d\rangle=0$ for every d. Taking $d=\nabla f(x^\star)$ gives $\|\nabla f(x^\star)\|^2=0$, i.e. $\nabla f(x^\star)=0$. □

2. Stability of GD on a quadratic $\frac{1}{2}x^\top Qx-b^\top x$, $Q\succ 0$: convergence from every x_0 $\iff$ $0<\alpha<2/\lambda_{\max}(Q)$.

Proof. For the error $e_k=x_k-x^\star$ (where $Qx^\star=b$): $e_{k+1}=e_k-\alpha Qe_k=(I-\alpha Q)e_k$. In the eigenbasis $Q=U\Lambda U^\top$ the coordinates $z=U^\top e$ evolve independently: $z_{k+1,i}=(1-\alpha\lambda_i)z_{k,i}$. Each coordinate tends to zero iff $|1-\alpha\lambda_i|<1$, i.e. $0<\alpha<2/\lambda_i$. The tightest restriction comes from $\lambda_{\max}$. □

Problem 1: type of a stationary point

Find the stationary points of $f(x,y)=5x^2+6xy+5y^2+10x+6y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x}=10x+6y+10=0,\quad \dfrac{\partial f}{\partial y}=6x+10y+6=0$. The system is linear with determinant $64\neq 0$, so the unique solution is $x^\star=(-1,\ 0)$.

2. Hessian. $H=\begin{pmatrix}10 & 6\\ 6 & 10\end{pmatrix}$ (constant). Sylvester's criterion: $h_{11}=10>0$, $\det H=10\cdot 10-6^2=64>0$, hence $H\succ 0$.

Via eigenvalues: $\lambda^2-20\lambda+64=0$, $\lambda_1=4,\ \lambda_2=16$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H(x - x^\star) > 0$ for $x \neq x^\star$, so the minimum is global and unique. $f(x^\star) = -5$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 4$, $L = \lambda_{\max}(H) = 16$, $\kappa = L/\mu = 4$. f is 4-strongly convex and 16-smooth. GD converges for $0 < \alpha < 2/L = \frac{1}{8}$; the standard step is $1/L = \frac{1}{16}$, the optimal step for a quadratic is $2/(L + \mu) = \frac{1}{10}$ with factor $(\kappa - 1)/(\kappa + 1) = \frac{3}{5}$.

Problem 2: gradient descent

For $f(x, y) = x^2 - xy + y^2$, perform three steps of gradient descent from $x_0 = (8,\ 4)$ with step $\alpha = \frac{1}{2}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (2x - y,\ -x + 2y)$, $H = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, eigenvalues $\mu = 1$, $L = 3$. Minimizer $x^\star = (0,\ 0)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{2} < 2/L = \frac{2}{3}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{2}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(8,\ 4)$	48	$(12,\ 0)$
1	$(2,\ 4)$	12	$(0,\ 6)$
2	$(2,\ 1)$	3	$(3,\ 0)$
3	$\left(\frac{1}{2},\ 1\right)$	$\frac{3}{4}$	—

Why it behaves this way. Eigendirections of H: $(1, 1)$ with $\lambda = 1$ and $(1, -1)$ with $\lambda = 3$. Each step multiplies the error components along them by $1 - \frac{1}{2} \cdot 1 = \frac{1}{2}$ and $1 - \frac{1}{2} \cdot 3 = -\frac{1}{2}$ respectively. A negative factor $-\frac{1}{2}$ causes overshooting past the minimizer (zig-zag), but $|-\frac{1}{2}| < 1$, so the oscillations die out. The overall speed is set by the largest $|1 - \alpha\lambda| = \frac{1}{2}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

Ticket 3

Definitions

1. **Directional derivative.** $D_d f(x) = \lim_{t\downarrow 0} \frac{f(x+td)-f(x)}{t}$. For differentiable f: $D_d f(x) = \langle \nabla f(x), d\rangle$.
2. **Hessian.** The matrix of second partial derivatives $\nabla^2 f(x) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right)_{i,j}$. It is symmetric when the mixed derivatives are continuous (Clairaut). The quantity $d^\top \nabla^2 f(x) d$ is the curvature along d.
3. **Feasible direction.** A vector d is feasible at $x \in \mathcal{X}$ if $x + td \in \mathcal{X}$ for all sufficiently small $t \geq 0$. Necessary condition for a minimum: $\langle \nabla f(x^\star), d\rangle \geq 0$ for all feasible d.
4. **Convex combination and convex set.** A convex combination is $\sum_i \lambda_i x_i$ with $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$. A set C is convex if $(1-t)x + ty \in C$ for all $x, y \in C$, $t \in [0,1]$ (it contains the segment between any two of its points).
5. **Strictly convex and μ-strongly convex function.** Strictly convex: the convexity inequality is strict for $x \neq y$, $t \in (0,1)$. μ-strongly convex ($\mu > 0$): $f((1-t)x+ty) \leq (1-t)f(x)+tf(y)-\frac{\mu}{2}t(1-t)\|x-y\|^2$; for differentiable f: $f(y) \geq f(x)+\langle \nabla f(x), y-x\rangle+\frac{\mu}{2}\|y-x\|^2$; for C^2 it suffices that $\nabla^2 f \succeq \mu I$. Example: x^4 is strictly but not strongly convex.
6. **Condition number.** For an L-smooth μ-strongly convex function, $\kappa = L/\mu \geq 1$; for a quadratic, $\kappa = \lambda_{\max}(Q)/\lambda_{\min}(Q)$. The larger κ, the more elongated the level sets and the slower gradient descent (contraction factor $1 - 1/\kappa$ for $\alpha = 1/L$).

Theoretical results

1. Second-order necessary condition: at an unconstrained local minimizer $\nabla^2 f(x^\star) \succeq 0$.
Proof. By the first-order condition, $\nabla f(x^\star) = 0$. Taylor expansion along td: $f(x^\star + td) - f(x^\star) = \frac{t^2}{2} d^\top \nabla^2 f(x^\star) d + o(t^2)$. The left side is ≥ 0 for small t. Dividing by $t^2/2$ and letting $t \to 0$ gives $d^\top \nabla^2 f(x^\star) d \geq 0$ for every d, i.e. $\nabla^2 f(x^\star) \succeq 0$. □

2. Strong convexity implies the Polyak–Lojasiewicz condition.
Proof. By μ-strong convexity, $f(y) \geq f(x)+\langle \nabla f(x), y-x\rangle+\frac{\mu}{2}\|y-x\|^2$ for all y. Minimizing the right side over y: the minimum is attained at $y = x - \frac{1}{\mu}\nabla f(x)$ and equals $f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2$. Since the left side at $y = x^\star$ equals $f^\star$, we get $f^\star \geq f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2$, i.e. $\frac{1}{2}\|\nabla f(x)\|^2 \geq \mu(f(x) - f^\star)$. □

Problem 1: type of a stationary point

Find the stationary points of $f(x,y) = 5x^2 + 2xy + 5y^2 - 30x - 6y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\frac{\partial f}{\partial x} = 10x + 2y - 30 = 0, \quad \frac{\partial f}{\partial y} = 2x + 10y - 6 = 0$. The system is linear with determinant $96 \neq 0$, so the unique solution is $x^\star = (3,\ 0)$.

2. Hessian. $H = \begin{pmatrix} 10 & 2 \\ 2 & 10 \end{pmatrix}$ (constant). Sylvester's criterion: $h_{11} = 10 > 0$, $\det H = 10 \cdot 10 - 2^2 = 96 > 0$, hence $H \succ 0$.

Via eigenvalues: $\lambda^2 - 20\lambda + 96 = 0$, $\lambda_1 = 8,\ \lambda_2 = 12$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H(x - x^\star) > 0$ for $x \neq x^\star$, so the minimum is global and unique. $f(x^\star) = -45$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 8$, $L = \lambda_{\max}(H) = 12$, $\kappa = L/\mu = \frac{3}{2}$. f is 8-strongly convex and 12-smooth. GD converges for $0 < \alpha < 2/L = \frac{1}{6}$; the standard step is $1/L = \frac{1}{12}$, the optimal step for a quadratic is $2/(L + \mu) = \frac{1}{10}$ with factor $(\kappa - 1)/(\kappa + 1) = \frac{1}{5}$.

Problem 2: gradient descent

For $f(x, y) = 2(x - 1)^2 + (y + 1)^2$, perform three steps of gradient descent from $x_0 = (10,\ 8)$ with step $\alpha = \frac{1}{3}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (4(x - 1),\ 2(y + 1))$, $H = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$, eigenvalues $\mu = 2$, $L = 4$. Minimizer $x^\star = (1,\ -1)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{3} < 2/L = \frac{1}{2}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{3}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(10,\ 8)$	243	$(36,\ 18)$
1	$(-2,\ 2)$	27	$(-12,\ 6)$
2	$(2,\ 0)$	3	$(4,\ 2)$
3	$\left(\frac{2}{3},\ -\frac{2}{3}\right)$	$\frac{1}{3}$	—

Why it behaves this way. Each step multiplies the deviation from the minimizer in each coordinate by $1 - \alpha\lambda$: in x by $1 - \frac{1}{3} \cdot 4 = -\frac{1}{3}$, in y by $1 - \frac{1}{3} \cdot 2 = \frac{1}{3}$. A negative factor $-\frac{1}{3}$ causes overshooting past the minimizer (zig-zag), but $|-\frac{1}{3}| < 1$, so the oscillations die out. The overall speed is set by the largest $|1 - \alpha\lambda| = \frac{1}{3}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{1}{3}\right)^2 = \frac{1}{9}$.

Ticket 4

Definitions

1. **Polyak–Lojasiewicz condition.** f satisfies PL with $\mu > 0$ if $\frac{1}{2}\|\nabla f(x)\|^2 \geq \mu\,(f(x) - f^\star)$ for all x. It requires neither convexity nor a unique minimizer (example: $f(x, y) = x^2$).
2. **Descent direction.** d is a strict descent direction at x if $\langle \nabla f(x), d\rangle < 0$. Then $f(x + \alpha d) < f(x)$ for all sufficiently small $\alpha > 0$. The negative gradient $-\nabla f(x)$ is the direction of steepest descent in the Euclidean norm.
3. **Gradient descent with a constant step.** $x_{k+1} = x_k - \alpha \nabla f(x_k)$, $\alpha > 0$. For L-smooth f, any step $0 < \alpha < 2/L$ guarantees decrease; the standard choice is $\alpha = 1/L$.
4. **Armijo condition (backtracking).** A step α is accepted if $f(x + \alpha d) \leq f(x) + c\alpha\langle \nabla f(x), d\rangle$, $c \in (0, 1)$ (typically $c = 10^{-4}$). Backtracking: start from α_0 and shrink $\alpha \leftarrow \beta\alpha$, $\beta \in (0, 1)$, until the condition holds.
5. **Active constraint.** An inequality $g_i(x) \leq 0$ is active at a feasible x if $g_i(x) = 0$. Inactive constraints impose no local restriction; by complementary slackness an inactive constraint has $\lambda_i^\star = 0$.
6. **Newton's method (Newton direction).** Minimizes the quadratic model $m(p) = f(x) + g^\top p + \frac{1}{2} p^\top H p$, $g = \nabla f(x)$, $H = \nabla^2 f(x)$: the direction p_N solves $H p_N = -g$, and $x^+ = x + p_N$ (in practice with damping). Near a nondegenerate minimizer it converges quadratically.

Theoretical results

1. Weak duality: $q(\lambda, \nu) \leq f(x)$ for every feasible x and $\lambda \geq 0$.
Proof. By definition of the infimum, $q(\lambda, \nu) = \inf_z \mathcal{L}(z, \lambda, \nu) \leq \mathcal{L}(x, \lambda, \nu)$. For feasible x and $\lambda \geq 0$: $\sum_i \lambda_i g_i(x) \leq 0$ and $\sum_j \nu_j h_j(x) = 0$, so $\mathcal{L}(x, \lambda, \nu) \leq f(x)$. Hence $q(\lambda, \nu) \leq f(x)$; taking the supremum on the left and the infimum on the right gives $d^\star \leq p^\star$. □

2. Linear convergence of GD under PL: with $\alpha = 1/L$, $f(x_k) - f^\star \leq (1 - \mu/L)^k (f(x_0) - f^\star)$.
Proof. By L-smoothness (descent lemma) with $\alpha = 1/L$: $f(x_{k+1}) \leq f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2$. By PL, $\|\nabla f(x_k)\|^2 \geq 2\mu(f(x_k) - f^\star)$. Therefore $f(x_{k+1}) \leq f(x_k) - \frac{\mu}{L}(f(x_k) - f^\star)$. Subtracting $f^\star$: $f(x_{k+1}) - f^\star \leq (1 - \frac{\mu}{L})(f(x_k) - f^\star)$; iterate. Convexity was never used. □

Problem 1: type of a stationary point

Find the stationary points of $f(x, y) = 5x^2 + 4xy + 2y^2 - 10x - 4y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x} = 10x + 4y - 10 = 0, \quad \dfrac{\partial f}{\partial y} = 4x + 4y - 4 = 0.$ The system is linear with determinant $24 \neq 0$, so the unique solution is $x^\star = (1,\ 0)$.

2. Hessian. $H = \begin{pmatrix} 10 & 4 \\ 4 & 4 \end{pmatrix}$ (constant). Sylvester's criterion: $h_{11} = 10 > 0$, $\det H = 10 \cdot 4 - 4^2 = 24 > 0$, hence $H \succ 0$.

Via eigenvalues: $\lambda^2 - 14\lambda + 24 = 0$, $\lambda_1 = 2,\ \lambda_2 = 12$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H (x - x^\star) > 0$ for $x \neq x^\star$, so the minimum is global and unique. $f(x^\star) = -5$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 2$, $L = \lambda_{\max}(H) = 12$, $\kappa = L/\mu = 6$. f is 2-strongly convex and 12-smooth. GD converges for $0 < \alpha < 2/L = \frac{1}{6}$; the standard step is $1/L = \frac{1}{12}$, the optimal step for a quadratic is $2/(L+\mu) = \frac{1}{7}$ with factor $(\kappa-1)/(\kappa+1) = \frac{5}{7}$.

Problem 2: gradient descent

For $f(x,y) = (x-1)^2 + (y+2)^2$, perform three steps of gradient descent from $x_0 = (9,\ 6)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (2(x-1),\ 2(y+2))$, $H = \begin{pmatrix} 2 & 0 \\ 0 & 2 \end{pmatrix}$, eigenvalues $\mu = 2$, $L = 2$. Minimizer $x^\star = (1,\ -2)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{4} < 2/L = 1$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{4}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(9,\ 6)$	128	$(16,\ 16)$
1	$(5,\ 2)$	32	$(8,\ 8)$
2	$(3,\ 0)$	8	$(4,\ 4)$
3	$(2,\ -1)$	2	—

Why it behaves this way. Each step multiplies the deviation from the minimizer in each coordinate by $1 - \alpha\lambda$: in x by $1 - \frac{1}{4}\cdot 2 = \frac{1}{2}$, in y by $1 - \frac{1}{4}\cdot 2 = \frac{1}{2}$. The overall speed is set by the largest $|1-\alpha\lambda| = \frac{1}{2}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

Ticket 5

Definitions

1. **Gradient (via the linear approximation).** For differentiable $f : \mathbb{R}^d \to \mathbb{R}$, the gradient $\nabla f(x)$ is the vector giving the linear part of the increment: $f(x + h) = f(x) + \langle \nabla f(x), h\rangle + o(\|h\|)$. Equivalently, $df = \langle \nabla f(x), dx\rangle$.
2. **Convex function.** f on a convex set C is convex if $f((1-t)x+ty) \le (1-t)f(x)+tf(y)$ for all $x, y \in C$, $t \in [0, 1]$. For differentiable f this is equivalent to $f(y) \ge f(x)+\langle \nabla f(x), y-x\rangle$, and for C^2 functions to $\nabla^2 f \succeq 0$.
3. **Condition number.** For an L-smooth μ-strongly convex function, $\kappa = L/\mu \ge 1$; for a quadratic, $\kappa = \lambda_{\max}(Q)/\lambda_{\min}(Q)$. The larger κ, the more elongated the level sets and the slower gradient descent (contraction factor $1 - 1/\kappa$ for $\alpha = 1/L$).
4. **Gradient descent with a constant step.** $x_{k+1} = x_k - \alpha \nabla f(x_k)$, $\alpha > 0$. For L-smooth f, any step $0 < \alpha < 2/L$ guarantees decrease; the standard choice is $\alpha = 1/L$.
5. **Armijo condition (backtracking).** A step α is accepted if $f(x + \alpha d) \le f(x) + c\alpha\langle \nabla f(x), d\rangle$, $c \in (0, 1)$ (typically $c = 10^{-4}$). Backtracking: start from α_0 and shrink $\alpha \leftarrow \beta\alpha$, $\beta \in (0, 1)$, until the condition holds.
6. **Karush–Kuhn–Tucker (KKT) conditions.** Stationarity $\nabla_x \mathcal{L}(x^\star, \lambda^\star, \nu^\star) = 0$; primal feasibility $g_i(x^\star) \le 0$, $h_j(x^\star) = 0$; dual feasibility $\lambda_i^\star \ge 0$; complementary slackness $\lambda_i^\star g_i(x^\star) = 0$. For convex f, g_i and affine h_j they are sufficient for global optimality.

Theoretical results

1. A symmetric $H \succeq 0$ if and only if all its eigenvalues are nonnegative.
Proof. By the spectral theorem $H = Q\Lambda Q^\top$ with orthogonal Q. For $z = Q^\top x$: $x^\top Hx = z^\top \Lambda z = \sum_i \lambda_i z_i^2$. If all $\lambda_i \ge 0$, the sum is nonnegative. Conversely, for an eigenvector q_i: $q_i^\top H q_i = \lambda_i \|q_i\|^2 \ge 0$, so $\lambda_i \ge 0$. The case $H \succ 0$ is analogous. □

2. For $f(x) = \frac{1}{2}x^\top Ax - b^\top x + c$, $A = A^\top \succ 0$: $\nabla f = Ax - b$, $\nabla^2 f = A$, and $x^\star = A^{-1}b$ is the unique global minimizer.
Proof. $df = \frac{1}{2}(dx^\top Ax + x^\top A\, dx) - b^\top dx = \langle Ax - b, dx\rangle$ (using symmetry of A), so $\nabla f = Ax - b$; differentiating once more gives $\nabla^2 f = A$. Stationarity: $Ax = b$, $x^\star = A^{-1}b$. Substituting $b = Ax^\star$ yields the identity $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top A(x - x^\star)$, which is > 0 for $x \ne x^\star$ since $A \succ 0$. □

Problem 1: type of a stationary point

Find the stationary points of $f(x, y) = 3x^2 + 2xy + 3y^2 - 10x - 14y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x} = 6x + 2y - 10 = 0$, $\dfrac{\partial f}{\partial y} = 2x + 6y - 14 = 0$. The system is linear with determinant $32 \ne 0$, so the unique solution is $x^\star = (1,\ 2)$.

2. Hessian. $H = \begin{pmatrix} 6 & 2 \\ 2 & 6 \end{pmatrix}$ (constant). Sylvester's criterion: $h_{11} = 6 > 0$, $\det H = 6 \cdot 6 - 2^2 = 32 > 0$, hence $H \succ 0$.

Via eigenvalues: $\lambda^2 - 12\lambda + 32 = 0$, $\lambda_1 = 4,\ \lambda_2 = 8$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H(x - x^\star) > 0$ for $x \ne x^\star$, so the minimum is global and unique. $f(x^\star) = -19$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 4$, $L = \lambda_{\max}(H) = 8$, $\kappa = L/\mu = 2$. f is 4-strongly convex and 8-smooth. GD converges for $0 < \alpha < 2/L = \frac{1}{4}$; the standard step is $1/L = \frac{1}{8}$, the optimal step for a quadratic is $2/(L+\mu) = \frac{1}{6}$ with factor $(\kappa-1)/(\kappa+1) = \frac{1}{3}$.

Problem 2: gradient descent

For $f(x,y) = 3x^2 + (y-1)^2$, perform three steps of gradient descent from $x_0 = (4,\ 9)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (6x,\ 2(y-1))$, $H = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$, eigenvalues $\mu = 2$, $L = 6$. Minimizer $x^\star = (0,\ 1)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{4} < 2/L = \frac{1}{3}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{4}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(4,\ 9)$	112	$(24,\ 16)$
1	$(-2,\ 5)$	28	$(-12,\ 8)$
2	$(1,\ 3)$	7	$(6,\ 4)$
3	$\left(-\frac{1}{2},\ 2\right)$	$\frac{7}{4}$	—

Why it behaves this way. Each step multiplies the deviation from the minimizer in each coordinate by $1 - \alpha\lambda$: in x by $1 - \frac{1}{4}\cdot 6 = -\frac{1}{2}$, in y by $1 - \frac{1}{4}\cdot 2 = \frac{1}{2}$. A negative factor $-\frac{1}{2}$ causes overshooting past the minimizer (zig-zag), but $|-\frac{1}{2}| < 1$, so the oscillations die out. The overall speed is set by the largest $|1-\alpha\lambda| = \frac{1}{2}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

Ticket 6

Definitions

1. **Strict local minimum.** Like a local minimum, but with strict inequality: $f(x^\star) < f(x)$ for all $x \in \mathcal{X} \cap B_r(x^\star)$, $x \neq x^\star$.
2. **Positive definite and positive semidefinite matrix.** Symmetric H: $H \succeq 0$ if $x^\top H x \geq 0$ for all x; $H \succ 0$ if $x^\top H x > 0$ for all $x \neq 0$. Equivalently, all eigenvalues are ≥ 0 (respectively > 0). For 2×2, Sylvester's criterion: $H \succ 0 \iff h_{11} > 0,\ \det H > 0$.
3. **Feasible direction.** A vector d is feasible at $x \in \mathcal{X}$ if $x + td \in \mathcal{X}$ for all sufficiently small $t \geq 0$. Necessary condition for a minimum: $\langle \nabla f(x^\star), d \rangle \geq 0$ for all feasible d.
4. **Convex combination and convex set.** A convex combination is $\sum_i \lambda_i x_i$ with $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$. A set C is convex if $(1-t)x + ty \in C$ for all $x, y \in C$, $t \in [0,1]$ (it contains the segment between any two of its points).
5. **L-smoothness (Lipschitz gradient).** $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$ for all x, y. For $f \in C^2$ it suffices that $\|\nabla^2 f(x)\|_2 \leq L$. Consequence (descent lemma): $f(y) \leq f(x) + \langle \nabla f(x), y - x \rangle + \frac{L}{2}\|y - x\|^2$. For a quadratic, $L = \lambda_{\max}(Q)$.
6. **Polyak–Lojasiewicz condition.** f satisfies PL with $\mu > 0$ if $\frac{1}{2}\|\nabla f(x)\|^2 \geq \mu\,(f(x) - f^\star)$ for all x. It requires neither convexity nor a unique minimizer (example: $f(x, y) = x^2$).

Theoretical results

1. The intersection of any family of convex sets is convex.
Proof. Let $C = \bigcap_\alpha C_\alpha$, $x, y \in C$, $t \in [0,1]$. Then $x, y \in C_\alpha$ for every α; by convexity of C_α, $(1-t)x + ty \in C_\alpha$ for every α, hence $(1-t)x + ty \in C$. □

2. Newton's method reaches the minimizer of a quadratic with $Q \succ 0$ in one step.
Proof. For $f(x) = \frac{1}{2}x^\top Q x - b^\top x + c$: $\nabla f(x) = Qx - b$, $\nabla^2 f = Q$. The Newton equation $Qp = -(Qx - b)$ gives $p = -x + Q^{-1}b = x^\star - x$. Hence $x^+ = x + p = x^\star$ from any starting point. □

Problem 1: type of a stationary point

Find the stationary points of $f(x, y) = 3x^2 + 4xy + 3y^2 + 8x + 12y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x} = 6x + 4y + 8 = 0, \quad \dfrac{\partial f}{\partial y} = 4x + 6y + 12 = 0.$ The system is linear with determinant $20 \neq 0$, so the unique solution is $x^\star = (0,\ -2)$.

2. Hessian. $H = \begin{pmatrix} 6 & 4 \\ 4 & 6 \end{pmatrix}$ (constant). Sylvester's criterion: $h_{11} = 6 > 0$, $\det H = 6 \cdot 6 - 4^2 = 20 > 0$, hence $H \succ 0$.

Via eigenvalues: $\lambda^2 - 12\lambda + 20 = 0,\ \lambda_1 = 2,\ \lambda_2 = 10$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H (x - x^\star) > 0$ for $x \neq x^\star$, so the minimum is global and unique. $f(x^\star) = -12$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top H h$; if it is positive in every direction, the function

increases wherever we move (a “bowl”). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 2$, $L = \lambda_{\max}(H) = 10$, $\kappa = L/\mu = 5$. f is 2-strongly convex and 10-smooth. GD converges for $0 < \alpha < 2/L = \frac{1}{5}$; the standard step is $1/L = \frac{1}{10}$, the optimal step for a quadratic is $2/(L+\mu) = \frac{1}{6}$ with factor $(\kappa-1)/(\kappa+1) = \frac{2}{3}$.

Problem 2: gradient descent

For $f(x,y) = (x-2)^2 + 3(y+1)^2$, perform three steps of gradient descent from $x_0 = (10,\ 3)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (2(x-2),\ 6(y+1))$, $H = \begin{pmatrix} 2 & 0 \\ 0 & 6 \end{pmatrix}$, eigenvalues $\mu = 2$, $L = 6$. Minimizer $x^\star = (2,\ -1)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{4} < 2/L = \frac{1}{3}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{4}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(10,\ 3)$	112	$(16,\ 24)$
1	$(6,\ -3)$	28	$(8,\ -12)$
2	$(4,\ 0)$	7	$(4,\ 6)$
3	$\left(3,\ -\frac{3}{2}\right)$	$\frac{7}{4}$	—

Why it behaves this way. Each step multiplies the deviation from the minimizer in each coordinate by $1 - \alpha\lambda$: in x by $1 - \frac{1}{4}\cdot 2 = \frac{1}{2}$, in y by $1 - \frac{1}{4}\cdot 6 = -\frac{1}{2}$. A negative factor $-\frac{1}{2}$ causes overshooting past the minimizer (zig-zag), but $|-\frac{1}{2}| < 1$, so the oscillations die out. The overall speed is set by the largest $|1-\alpha\lambda| = \frac{1}{2}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

Ticket 7

Definitions

1. **Directional derivative.** $D_d f(x) = \lim_{t\downarrow 0} \frac{f(x+td)-f(x)}{t}$. For differentiable f: $D_d f(x) = \langle \nabla f(x), d\rangle$.
2. **Hessian.** The matrix of second partial derivatives $\nabla^2 f(x) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right)_{i,j}$. It is symmetric when the mixed derivatives are continuous (Clairaut). The quantity $d^\top \nabla^2 f(x) d$ is the curvature along d.
3. **Stationary point and saddle point.** A stationary point satisfies $\nabla f(x) = 0$. A saddle point is a stationary point that is neither a local minimum nor a local maximum (there are directions of both increase and decrease; e.g. an indefinite Hessian, as for $x^2 - y^2$ at the origin).
4. **Strictly convex and μ-strongly convex function.** Strictly convex: the convexity inequality is strict for $x \neq y$, $t \in (0,1)$. μ-strongly convex ($\mu > 0$): $f((1-t)x+ty) \leq (1-t)f(x)+tf(y)-\frac{\mu}{2}t(1-t)\|x-y\|^2$; for differentiable f: $f(y) \geq f(x)+\langle \nabla f(x), y-x\rangle+\frac{\mu}{2}\|y-x\|^2$; for C^2 it suffices that $\nabla^2 f \succeq \mu I$. Example: x^4 is strictly but not strongly convex.
5. **Active constraint.** An inequality $g_i(x) \leq 0$ is active at a feasible x if $g_i(x) = 0$. Inactive constraints impose no local restriction; by complementary slackness an inactive constraint has $\lambda_i^\star = 0$.
6. **Dual function and dual problem.** $q(\lambda,\nu) = \inf_x \mathcal{L}(x,\lambda,\nu)$ is always concave (an infimum of affine functions). Dual problem: $\max_{\lambda\geq 0,\nu} q(\lambda,\nu)$. Weak duality: $d^\star \leq p^\star$.

Theoretical results

1. For $f(x) = \frac{1}{2}x^\top Ax - b^\top x + c$, $A = A^\top \succ 0$: $\nabla f = Ax - b$, $\nabla^2 f = A$, and $x^\star = A^{-1}b$ is the unique global minimizer.
Proof. $df = \frac{1}{2}(dx^\top Ax + x^\top A\,dx) - b^\top dx = \langle Ax - b, dx\rangle$ (using symmetry of A), so $\nabla f = Ax - b$; differentiating once more gives $\nabla^2 f = A$. Stationarity: $Ax = b$, $x^\star = A^{-1}b$. Substituting $b = Ax^\star$ yields the identity $f(x) - f(x^\star) = \frac{1}{2}(x-x^\star)^\top A(x-x^\star)$, which is > 0 for $x \neq x^\star$ since $A \succ 0$. □

2. The negative gradient is the direction of steepest local decrease (among $\|d\|_2 = 1$).
Proof. The rate of change along a unit d is $D_d f(x) = \langle \nabla f(x), d\rangle$. By Cauchy–Schwarz, $\langle \nabla f(x), d\rangle \geq -\|\nabla f(x)\|\|d\| = -\|\nabla f(x)\|$, with equality for $d = -\nabla f(x)/\|\nabla f(x)\|$. So this direction gives the smallest (most negative) derivative. □

Problem 1: type of a stationary point

Find the stationary points of $f(x,y) = 3x^2 + 4xy + 6y^2 + 8x - 4y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\frac{\partial f}{\partial x} = 6x + 4y + 8 = 0, \quad \frac{\partial f}{\partial y} = 4x + 12y - 4 = 0.$ The system is linear with determinant $56 \neq 0$, so the unique solution is $x^\star = (-2,\ 1)$.

2. Hessian. $H = \begin{pmatrix} 6 & 4 \\ 4 & 12 \end{pmatrix}$ (constant). Sylvester's criterion: $h_{11} = 6 > 0$, $\det H = 6\cdot 12 - 4^2 = 56 > 0$, hence $H \succ 0$.

Via eigenvalues: $\lambda^2 - 18\lambda + 56 = 0$, $\lambda_1 = 4,\ \lambda_2 = 14$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H(x - x^\star) > 0$ for $x \neq x^\star$, so the minimum is global and unique. $f(x^\star) = -10$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 4$, $L = \lambda_{\max}(H) = 14$, $\kappa = L/\mu = \frac{7}{2}$. f is 4-strongly convex and 14-smooth. GD converges for $0 < \alpha < 2/L = \frac{1}{7}$; the standard step is $1/L = \frac{1}{14}$, the optimal step for a quadratic is $2/(L + \mu) = \frac{1}{9}$ with factor $(\kappa - 1)/(\kappa + 1) = \frac{5}{9}$.

Problem 2: gradient descent

For $f(x, y) = 2x^2 + 2y^2$, perform three steps of gradient descent from $x_0 = (6,\ -4)$ with step $\alpha = \frac{1}{8}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (4x,\ 4y)$, $H = \begin{pmatrix} 4 & 0 \\ 0 & 4 \end{pmatrix}$, eigenvalues $\mu = 4$, $L = 4$. Minimizer $x^\star = (0,\ 0)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{8} < 2/L = \frac{1}{2}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{8}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(6,\ -4)$	104	$(24,\ -16)$
1	$(3,\ -2)$	26	$(12,\ -8)$
2	$\left(\frac{3}{2},\ -1\right)$	$\frac{13}{2}$	$(6,\ -4)$
3	$\left(\frac{3}{4},\ -\frac{1}{2}\right)$	$\frac{13}{8}$	—

Why it behaves this way. Each step multiplies the deviation from the minimizer in each coordinate by $1 - \alpha\lambda$: in x by $1 - \frac{1}{8} \cdot 4 = \frac{1}{2}$, in y by $1 - \frac{1}{8} \cdot 4 = \frac{1}{2}$. The overall speed is set by the largest $|1 - \alpha\lambda| = \frac{1}{2}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

Ticket 8

Definitions

1. **Optimization problem: decision variable, objective, feasible set.** A problem $\min_{x\in\mathcal{X}} f(x)$, where x is the decision variable (e.g. model weights), f is the objective (error, cost, risk), and $\mathcal{X}$ is the feasible set (constraints; without constraints $\mathcal{X}=\mathbb{R}^d$). Maximizing f is equivalent to minimizing $-f$.
2. **Local minimum.** $x^\star\in\mathcal{X}$ is a local minimizer if there exists $r>0$ such that $f(x^\star)\le f(x)$ for all $x\in\mathcal{X}\cap B_r(x^\star)$.
3. **Global minimum.** $x^\star\in\mathcal{X}$ is a global minimizer if $f(x^\star)\le f(x)$ for all $x\in\mathcal{X}$. A minimum may fail to exist (e^x on $\mathbb{R}$) and may be non-unique ($(x^2-1)^2$).
4. **Descent direction.** d is a strict descent direction at x if $\langle\nabla f(x),d\rangle<0$. Then $f(x+\alpha d)<f(x)$ for all sufficiently small $\alpha>0$. The negative gradient $-\nabla f(x)$ is the direction of steepest descent in the Euclidean norm.
5. **Lagrangian.** For $\min f(x)$ subject to $g_i(x)\le 0$, $h_j(x)=0$: $\mathcal{L}(x,\lambda,\nu)=f(x)+\sum_i\lambda_i g_i(x)+\sum_j\nu_j h_j(x)$, where $\lambda_i\ge 0$ and ν_j are unrestricted in sign.
6. **Newton's method (Newton direction).** Minimizes the quadratic model $m(p)=f(x)+g^\top p+\frac{1}{2}p^\top Hp$, $g=\nabla f(x)$, $H=\nabla^2 f(x)$: the direction p_N solves $Hp_N=-g$, and $x^+=x+p_N$ (in practice with damping). Near a nondegenerate minimizer it converges quadratically.

Theoretical results

1. Second-order necessary condition: at an unconstrained local minimizer $\nabla^2 f(x^\star)\succeq 0$.
Proof. By the first-order condition, $\nabla f(x^\star)=0$. Taylor expansion along td: $f(x^\star+td)-f(x^\star)=\frac{t^2}{2}d^\top\nabla^2 f(x^\star)d+o(t^2)$. The left side is ≥ 0 for small t. Dividing by $t^2/2$ and letting $t\to 0$ gives $d^\top\nabla^2 f(x^\star)d\ge 0$ for every d, i.e. $\nabla^2 f(x^\star)\succeq 0$. □

2. Linear convergence of GD under PL: with $\alpha=1/L$, $f(x_k)-f^\star\le(1-\mu/L)^k(f(x_0)-f^\star)$.
Proof. By L-smoothness (descent lemma) with $\alpha=1/L$: $f(x_{k+1})\le f(x_k)-\frac{1}{2L}\|\nabla f(x_k)\|^2$. By PL, $\|\nabla f(x_k)\|^2\ge 2\mu(f(x_k)-f^\star)$. Therefore $f(x_{k+1})\le f(x_k)-\frac{\mu}{L}(f(x_k)-f^\star)$. Subtracting $f^\star$: $f(x_{k+1})-f^\star\le(1-\frac{\mu}{L})(f(x_k)-f^\star)$; iterate. Convexity was never used. □

Problem 1: type of a stationary point

Find the stationary points of $f(x,y)=4x^2-4xy+4y^2-16x+20y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x}=8x-4y-16=0,\quad \dfrac{\partial f}{\partial y}=-4x+8y+20=0$. The system is linear with determinant $48\neq 0$, so the unique solution is $x^\star=(1,\ -2)$.

2. Hessian. $H=\begin{pmatrix}8&-4\\-4&8\end{pmatrix}$ (constant). Sylvester's criterion: $h_{11}=8>0$, $\det H=8\cdot 8-4^2=48>0$, hence $H\succ 0$.

Via eigenvalues: $\lambda^2-16\lambda+48=0$, $\lambda_1=4,\ \lambda_2=12$ — both positive.

3. Conclusion. $\nabla f(x^\star)=0$ and $H\succ 0\Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x)-f(x^\star)=\frac{1}{2}(x-x^\star)^\top H(x-x^\star)>0$ for $x\neq x^\star$, so the minimum is global and unique. $f(x^\star)=-28$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 4$, $L = \lambda_{\max}(H) = 12$, $\kappa = L/\mu = 3$. f is 4-strongly convex and 12-smooth. GD converges for $0 < \alpha < 2/L = \frac{1}{6}$; the standard step is $1/L = \frac{1}{12}$, the optimal step for a quadratic is $2/(L+\mu) = \frac{1}{8}$ with factor $(\kappa-1)/(\kappa+1) = \frac{1}{2}$.

Problem 2: gradient descent

For $f(x,y) = 2(x-2)^2 + (y-1)^2$, perform three steps of gradient descent from $x_0 = (1,\ 9)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (4(x-2),\ 2(y-1))$, $H = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$, eigenvalues $\mu = 2$, $L = 4$. Minimizer $x^\star = (2,\ 1)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{4} < 2/L = \frac{1}{2}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{4}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(1,\ 9)$	66	$(-4,\ 16)$
1	$(2,\ 5)$	16	$(0,\ 8)$
2	$(2,\ 3)$	4	$(0,\ 4)$
3	$(2,\ 2)$	1	—

Why it behaves this way. Each step multiplies the deviation from the minimizer in each coordinate by $1 - \alpha\lambda$: in x by $1 - \frac{1}{4}\cdot 4 = 0$, in y by $1 - \frac{1}{4}\cdot 2 = \frac{1}{2}$. A factor of 0 means this component is eliminated in a single step (step $\alpha = 1/\lambda$). The overall speed is set by the largest $|1-\alpha\lambda| = \frac{1}{2}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

Ticket 9

Definitions

1. **Strict local minimum.** Like a local minimum, but with strict inequality: $f(x^\star) < f(x)$ for all $x \in \mathcal{X} \cap B_r(x^\star)$, $x \neq x^\star$.
2. **Directional derivative.** $D_d f(x) = \lim_{t \downarrow 0} \frac{f(x+td)-f(x)}{t}$. For differentiable f: $D_d f(x) = \langle \nabla f(x), d \rangle$.
3. **Strictly convex and μ-strongly convex function.** Strictly convex: the convexity inequality is strict for $x \neq y$, $t \in (0,1)$. μ-strongly convex ($\mu > 0$): $f((1-t)x+ty) \leq (1-t)f(x)+tf(y)-\frac{\mu}{2}t(1-t)\|x-y\|^2$; for differentiable f: $f(y) \geq f(x)+\langle \nabla f(x), y-x\rangle+\frac{\mu}{2}\|y-x\|^2$; for C^2 it suffices that $\nabla^2 f \succeq \mu I$. Example: x^4 is strictly but not strongly convex.
4. **Condition number.** For an L-smooth μ-strongly convex function, $\kappa = L/\mu \geq 1$; for a quadratic, $\kappa = \lambda_{\max}(Q)/\lambda_{\min}(Q)$. The larger κ, the more elongated the level sets and the slower gradient descent (contraction factor $1 - 1/\kappa$ for $\alpha = 1/L$).
5. **Armijo condition (backtracking).** A step α is accepted if $f(x + \alpha d) \leq f(x) + c\alpha\langle \nabla f(x), d\rangle$, $c \in (0,1)$ (typically $c = 10^{-4}$). Backtracking: start from α_0 and shrink $\alpha \leftarrow \beta\alpha$, $\beta \in (0,1)$, until the condition holds.
6. **Dual function and dual problem.** $q(\lambda,\nu) = \inf_x \mathcal{L}(x,\lambda,\nu)$ is always concave (an infimum of affine functions). Dual problem: $\max_{\lambda \geq 0, \nu} q(\lambda,\nu)$. Weak duality: $d^\star \leq p^\star$.

Theoretical results

1. Strong convexity implies the Polyak–Lojasiewicz condition.
Proof. By μ-strong convexity, $f(y) \geq f(x)+\langle \nabla f(x), y-x\rangle+\frac{\mu}{2}\|y-x\|^2$ for all y. Minimizing the right side over y: the minimum is attained at $y = x - \frac{1}{\mu}\nabla f(x)$ and equals $f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2$. Since the left side at $y = x^\star$ equals $f^\star$, we get $f^\star \geq f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2$, i.e. $\frac{1}{2}\|\nabla f(x)\|^2 \geq \mu(f(x) - f^\star)$. □

2. Stability of GD on a quadratic $\frac{1}{2}x^\top Qx - b^\top x$, $Q \succ 0$: convergence from every x_0 $\iff$ $0 < \alpha < 2/\lambda_{\max}(Q)$.
Proof. For the error $e_k = x_k - x^\star$ (where $Qx^\star = b$): $e_{k+1} = e_k - \alpha Q e_k = (I - \alpha Q)e_k$. In the eigenbasis $Q = U\Lambda U^\top$ the coordinates $z = U^\top e$ evolve independently: $z_{k+1,i} = (1 - \alpha\lambda_i)z_{k,i}$. Each coordinate tends to zero iff $|1 - \alpha\lambda_i| < 1$, i.e. $0 < \alpha < 2/\lambda_i$. The tightest restriction comes from $\lambda_{\max}$. □

Problem 1: type of a stationary point

Find the stationary points of $f(x,y) = 2x^2 - xy + 2y^2 - 9x + 6y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x} = 4x - y - 9 = 0$, $\dfrac{\partial f}{\partial y} = -x + 4y + 6 = 0$. The system is linear with determinant $15 \neq 0$, so the unique solution is $x^\star = (2,\ -1)$.

2. Hessian. $H = \begin{pmatrix} 4 & -1 \\ -1 & 4 \end{pmatrix}$ (constant). Sylvester's criterion: $h_{11} = 4 > 0$, $\det H = 4 \cdot 4 - 1^2 = 15 > 0$, hence $H \succ 0$.

Via eigenvalues: $\lambda^2 - 8\lambda + 15 = 0$, $\lambda_1 = 3,\ \lambda_2 = 5$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H(x - x^\star) > 0$ for $x \neq x^\star$, so the minimum is global and unique.

$f(x^\star) = -12$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top H h$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 3$, $L = \lambda_{\max}(H) = 5$, $\kappa = L/\mu = \frac{5}{3}$. f is 3-strongly convex and 5-smooth. GD converges for $0 < \alpha < 2/L = \frac{2}{5}$; the standard step is $1/L = \frac{1}{5}$, the optimal step for a quadratic is $2/(L+\mu) = \frac{1}{4}$ with factor $(\kappa-1)/(\kappa+1) = \frac{1}{4}$.

Problem 2: gradient descent

For $f(x,y) = x^2 - xy + y^2$, perform three steps of gradient descent from $x_0 = (4,\ 0)$ with step $\alpha = \frac{1}{2}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (2x - y,\ -x + 2y)$, $H = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$, eigenvalues $\mu = 1$, $L = 3$. Minimizer $x^\star = (0,\ 0)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{2} < 2/L = \frac{2}{3}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{2}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(4,\ 0)$	16	$(8,\ -4)$
1	$(0,\ 2)$	4	$(-2,\ 4)$
2	$(1,\ 0)$	1	$(2,\ -1)$
3	$\left(0,\ \frac{1}{2}\right)$	$\frac{1}{4}$	—

Why it behaves this way. Eigendirections of H: $(1,1)$ with $\lambda = 1$ and $(1,-1)$ with $\lambda = 3$. Each step multiplies the error components along them by $1 - \frac{1}{2}\cdot 1 = \frac{1}{2}$ and $1 - \frac{1}{2}\cdot 3 = -\frac{1}{2}$ respectively. A negative factor $-\frac{1}{2}$ causes overshooting past the minimizer (zig-zag), but $|-\frac{1}{2}| < 1$, so the oscillations die out. The overall speed is set by the largest $|1 - \alpha\lambda| = \frac{1}{2}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

Ticket 10

Definitions

1. **Local minimum.** $x^\star \in \mathcal{X}$ is a local minimizer if there exists $r > 0$ such that $f(x^\star) \le f(x)$ for all $x \in \mathcal{X} \cap B_r(x^\star)$.
2. **Hessian.** The matrix of second partial derivatives $\nabla^2 f(x) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right)_{i,j}$. It is symmetric when the mixed derivatives are continuous (Clairaut). The quantity $d^\top \nabla^2 f(x) d$ is the curvature along d.
3. **Feasible direction.** A vector d is feasible at $x \in \mathcal{X}$ if $x + td \in \mathcal{X}$ for all sufficiently small $t \ge 0$. Necessary condition for a minimum: $\langle \nabla f(x^\star), d\rangle \ge 0$ for all feasible d.
4. **Convex function.** f on a convex set C is convex if $f((1-t)x+ty) \le (1-t)f(x)+tf(y)$ for all $x, y \in C$, $t \in [0,1]$. For differentiable f this is equivalent to $f(y) \ge f(x)+\langle \nabla f(x), y-x\rangle$, and for C^2 functions to $\nabla^2 f \succeq 0$.
5. **Descent direction.** d is a strict descent direction at x if $\langle \nabla f(x), d\rangle < 0$. Then $f(x + \alpha d) < f(x)$ for all sufficiently small $\alpha > 0$. The negative gradient $-\nabla f(x)$ is the direction of steepest descent in the Euclidean norm.
6. **Newton's method (Newton direction).** Minimizes the quadratic model $m(p) = f(x) + g^\top p + \frac{1}{2} p^\top H p$, $g = \nabla f(x)$, $H = \nabla^2 f(x)$: the direction p_N solves $H p_N = -g$, and $x^+ = x + p_N$ (in practice with damping). Near a nondegenerate minimizer it converges quadratically.

Theoretical results

1. A symmetric $H \succeq 0$ if and only if all its eigenvalues are nonnegative.
Proof. By the spectral theorem $H = Q\Lambda Q^\top$ with orthogonal Q. For $z = Q^\top x$: $x^\top H x = z^\top \Lambda z = \sum_i \lambda_i z_i^2$. If all $\lambda_i \ge 0$, the sum is nonnegative. Conversely, for an eigenvector q_i: $q_i^\top H q_i = \lambda_i \|q_i\|^2 \ge 0$, so $\lambda_i \ge 0$. The case $H \succ 0$ is analogous. □

2. Guaranteed decrease of a GD step for L-smooth f: $f(x^+) \le f(x) - \alpha(1-\frac{L\alpha}{2})\|\nabla f(x)\|^2$.
Proof. Descent lemma (may be used without proof): $f(y) \le f(x) + \langle \nabla f(x), y - x\rangle + \frac{L}{2}\|y - x\|^2$. Substitute $y = x^+ = x - \alpha \nabla f(x)$: $f(x^+) \le f(x) - \alpha\|\nabla f(x)\|^2 + \frac{L\alpha^2}{2}\|\nabla f(x)\|^2 = f(x) - \alpha\left(1 - \frac{L\alpha}{2}\right)\|\nabla f(x)\|^2$. Hence for $0 < \alpha < 2/L$ there is strict decrease away from stationary points, and for $\alpha = 1/L$: $f(x^+) \le f(x) - \frac{1}{2L}\|\nabla f(x)\|^2$. □

Problem 1: type of a stationary point

Find the stationary points of $f(x,y) = 3x^2 + 3xy + 3y^2 + 9x + 9y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x} = 6x + 3y + 9 = 0, \quad \dfrac{\partial f}{\partial y} = 3x + 6y + 9 = 0.$ The system is linear with determinant $27 \neq 0$, so the unique solution is $x^\star = (-1,\ -1)$.

2. Hessian. $H = \begin{pmatrix} 6 & 3 \\ 3 & 6 \end{pmatrix}$ (constant). Sylvester's criterion: $h_{11} = 6 > 0$, $\det H = 6 \cdot 6 - 3^2 = 27 > 0$, hence $H \succ 0$.

Via eigenvalues: $\lambda^2 - 12\lambda + 27 = 0$, $\lambda_1 = 3,\ \lambda_2 = 9$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H (x - x^\star) > 0$ for $x \neq x^\star$, so the minimum is global and unique. $f(x^\star) = -9$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 3$, $L = \lambda_{\max}(H) = 9$, $\kappa = L/\mu = 3$. f is 3-strongly convex and 9-smooth. GD converges for $0 < \alpha < 2/L = \frac{2}{9}$; the standard step is $1/L = \frac{1}{9}$, the optimal step for a quadratic is $2/(L+\mu) = \frac{1}{6}$ with factor $(\kappa-1)/(\kappa+1) = \frac{1}{2}$.

Problem 2: gradient descent

For $f(x,y) = (x-1)^2 + 2(y+1)^2$, perform three steps of gradient descent from $x_0 = (5,\ 3)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (2(x-1),\ 4(y+1))$, $H = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$, eigenvalues $\mu = 2$, $L = 4$. Minimizer $x^\star = (1,\ -1)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{4} < 2/L = \frac{1}{2}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{4}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(5,\ 3)$	48	$(8,\ 16)$
1	$(3,\ -1)$	4	$(4,\ 0)$
2	$(2,\ -1)$	1	$(2,\ 0)$
3	$\left(\frac{3}{2},\ -1\right)$	$\frac{1}{4}$	—

Why it behaves this way. Each step multiplies the deviation from the minimizer in each coordinate by $1 - \alpha\lambda$: in x by $1 - \frac{1}{4}\cdot 2 = \frac{1}{2}$, in y by $1 - \frac{1}{4}\cdot 4 = 0$. A factor of 0 means this component is eliminated in a single step (step $\alpha = 1/\lambda$). The overall speed is set by the largest $|1-\alpha\lambda| = \frac{1}{2}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

Ticket 11

Definitions

1. **Optimization problem: decision variable, objective, feasible set.** A problem $\min_{x\in\mathcal{X}} f(x)$, where x is the decision variable (e.g. model weights), f is the objective (error, cost, risk), and $\mathcal{X}$ is the feasible set (constraints; without constraints $\mathcal{X}=\mathbb{R}^d$). Maximizing f is equivalent to minimizing $-f$.
2. **Gradient (via the linear approximation).** For differentiable $f:\mathbb{R}^d\to\mathbb{R}$, the gradient $\nabla f(x)$ is the vector giving the linear part of the increment: $f(x+h)=f(x)+\langle\nabla f(x),h\rangle+o(\|h\|)$. Equivalently, $df=\langle\nabla f(x),dx\rangle$.
3. **Stationary point and saddle point.** A stationary point satisfies $\nabla f(x)=0$. A saddle point is a stationary point that is neither a local minimum nor a local maximum (there are directions of both increase and decrease; e.g. an indefinite Hessian, as for x^2-y^2 at the origin).
4. **Convex combination and convex set.** A convex combination is $\sum_i \lambda_i x_i$ with $\lambda_i\geq 0$, $\sum_i\lambda_i=1$. A set C is convex if $(1-t)x+ty\in C$ for all $x,y\in C$, $t\in[0,1]$ (it contains the segment between any two of its points).
5. **Lagrangian.** For $\min f(x)$ subject to $g_i(x)\leq 0$, $h_j(x)=0$: $\mathcal{L}(x,\lambda,\nu)=f(x)+\sum_i\lambda_i g_i(x)+\sum_j\nu_j h_j(x)$, where $\lambda_i\geq 0$ and ν_j are unrestricted in sign.
6. **Active constraint.** An inequality $g_i(x)\leq 0$ is active at a feasible x if $g_i(x)=0$. Inactive constraints impose no local restriction; by complementary slackness an inactive constraint has $\lambda_i^\star=0$.

Theoretical results

1. First-order necessary condition: at an unconstrained local minimizer $\nabla f(x^\star)=0$.

Proof. Fix a direction d and let $\phi(t)=f(x^\star+td)$. Since $x^\star$ is a local minimizer of f, $t=0$ is a local minimizer of the one-variable function ϕ, so $\phi'(0)=0$. By the chain rule, $\phi'(0)=\langle\nabla f(x^\star),d\rangle=0$ for every d. Taking $d=\nabla f(x^\star)$ gives $\|\nabla f(x^\star)\|^2=0$, i.e. $\nabla f(x^\star)=0$. □

2. Weak duality: $q(\lambda,\nu)\leq f(x)$ for every feasible x and $\lambda\geq 0$.

Proof. By definition of the infimum, $q(\lambda,\nu)=\inf_z\mathcal{L}(z,\lambda,\nu)\leq\mathcal{L}(x,\lambda,\nu)$. For feasible x and $\lambda\geq 0$: $\sum_i\lambda_i g_i(x)\leq 0$ and $\sum_j\nu_j h_j(x)=0$, so $\mathcal{L}(x,\lambda,\nu)\leq f(x)$. Hence $q(\lambda,\nu)\leq f(x)$; taking the supremum on the left and the infimum on the right gives $d^\star\leq p^\star$. □

Problem 1: type of a stationary point

Find the stationary points of $f(x,y)=x^2+xy+y^2-x+y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x}=2x+y-1=0,\quad \dfrac{\partial f}{\partial y}=x+2y+1=0.$ The system is linear with determinant $3\neq 0$, so the unique solution is $x^\star=(1,\,-1)$.

2. Hessian. $H=\begin{pmatrix}2&1\\1&2\end{pmatrix}$ (constant). Sylvester's criterion: $h_{11}=2>0$, $\det H=2\cdot 2-1^2=3>0$, hence $H\succ 0$.

Via eigenvalues: $\lambda^2-4\lambda+3=0$, $\lambda_1=1,\ \lambda_2=3$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H(x - x^\star) > 0$ for $x \neq x^\star$, so the minimum is global and unique. $f(x^\star) = -1$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 1$, $L = \lambda_{\max}(H) = 3$, $\kappa = L/\mu = 3$. f is 1-strongly convex and 3-smooth. GD converges for $0 < \alpha < 2/L = \frac{2}{3}$; the standard step is $1/L = \frac{1}{3}$, the optimal step for a quadratic is $2/(L + \mu) = \frac{1}{2}$ with factor $(\kappa - 1)/(\kappa + 1) = \frac{1}{2}$.

Problem 2: gradient descent

For $f(x, y) = 4(x - 1)^2 + (y - 1)^2$, perform three steps of gradient descent from $x_0 = (3,\ 17)$ with step $\alpha = \frac{1}{8}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (8(x - 1),\ 2(y - 1))$, $H = \begin{pmatrix} 8 & 0 \\ 0 & 2 \end{pmatrix}$, eigenvalues $\mu = 2$, $L = 8$. Minimizer $x^\star = (1,\ 1)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{8} < 2/L = \frac{1}{4}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{8}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(3,\ 17)$	272	$(16,\ 32)$
1	$(1,\ 13)$	144	$(0,\ 24)$
2	$(1,\ 10)$	81	$(0,\ 18)$
3	$\left(1,\ \frac{31}{4}\right)$	$\frac{729}{16}$	—

Why it behaves this way. Each step multiplies the deviation from the minimizer in each coordinate by $1 - \alpha\lambda$: in x by $1 - \frac{1}{8} \cdot 8 = 0$, in y by $1 - \frac{1}{8} \cdot 2 = \frac{3}{4}$. A factor of 0 means this component is eliminated in a single step (step $\alpha = 1/\lambda$). The overall speed is set by the largest $|1 - \alpha\lambda| = \frac{3}{4}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{3}{4}\right)^2 = \frac{9}{16}$.

Ticket 12

Definitions

1. **Global minimum.** $x^\star \in \mathcal{X}$ is a global minimizer if $f(x^\star) \le f(x)$ for all $x \in \mathcal{X}$. A minimum may fail to exist (e^x on $\mathbb{R}$) and may be non-unique ($(x^2-1)^2$).
2. **Positive definite and positive semidefinite matrix.** Symmetric H: $H \succeq 0$ if $x^\top Hx \ge 0$ for all x; $H \succ 0$ if $x^\top Hx > 0$ for all $x \ne 0$. Equivalently, all eigenvalues are ≥ 0 (respectively > 0). For 2×2, Sylvester's criterion: $H \succ 0 \iff h_{11} > 0,\ \det H > 0$.
3. **L-smoothness (Lipschitz gradient).** $\|\nabla f(x) - \nabla f(y)\| \le L\|x-y\|$ for all x, y. For $f \in C^2$ it suffices that $\|\nabla^2 f(x)\|_2 \le L$. Consequence (descent lemma): $f(y) \le f(x) + \langle \nabla f(x), y-x\rangle + \frac{L}{2}\|y-x\|^2$. For a quadratic, $L = \lambda_{\max}(Q)$.
4. **Polyak–Lojasiewicz condition.** f satisfies PL with $\mu > 0$ if $\frac{1}{2}\|\nabla f(x)\|^2 \ge \mu\,(f(x) - f^\star)$ for all x. It requires neither convexity nor a unique minimizer (example: $f(x,y) = x^2$).
5. **Gradient descent with a constant step.** $x_{k+1} = x_k - \alpha \nabla f(x_k)$, $\alpha > 0$. For L-smooth f, any step $0 < \alpha < 2/L$ guarantees decrease; the standard choice is $\alpha = 1/L$.
6. **Karush–Kuhn–Tucker (KKT) conditions.** Stationarity $\nabla_x \mathcal{L}(x^\star, \lambda^\star, \nu^\star) = 0$; primal feasibility $g_i(x^\star) \le 0$, $h_j(x^\star) = 0$; dual feasibility $\lambda_i^\star \ge 0$; complementary slackness $\lambda_i^\star g_i(x^\star) = 0$. For convex f, g_i and affine h_j they are sufficient for global optimality.

Theoretical results

1. The intersection of any family of convex sets is convex.
Proof. Let $C = \bigcap_\alpha C_\alpha$, $x, y \in C$, $t \in [0,1]$. Then $x, y \in C_\alpha$ for every α; by convexity of C_α, $(1-t)x + ty \in C_\alpha$ for every α, hence $(1-t)x + ty \in C$. □

2. Newton's method reaches the minimizer of a quadratic with $Q \succ 0$ in one step.
Proof. For $f(x) = \frac{1}{2}x^\top Qx - b^\top x + c$: $\nabla f(x) = Qx - b$, $\nabla^2 f = Q$. The Newton equation $Qp = -(Qx - b)$ gives $p = -x + Q^{-1}b = x^\star - x$. Hence $x^+ = x + p = x^\star$ from any starting point. □

Problem 1: type of a stationary point

Find the stationary points of $f(x,y) = 2x^2 + 4xy + 5y^2 - 4x - 16y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x} = 4x + 4y - 4 = 0, \quad \dfrac{\partial f}{\partial y} = 4x + 10y - 16 = 0.$ The system is linear with determinant $24 \ne 0$, so the unique solution is $x^\star = (-1,\ 2)$.

2. Hessian. $H = \begin{pmatrix} 4 & 4 \\ 4 & 10 \end{pmatrix}$ (constant). Sylvester's criterion: $h_{11} = 4 > 0$, $\det H = 4\cdot 10 - 4^2 = 24 > 0$, hence $H \succ 0$.

Via eigenvalues: $\lambda^2 - 14\lambda + 24 = 0,\ \lambda_1 = 2,\ \lambda_2 = 12$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H(x - x^\star) > 0$ for $x \ne x^\star$, so the minimum is global and unique. $f(x^\star) = -14$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function

increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 2$, $L = \lambda_{\max}(H) = 12$, $\kappa = L/\mu = 6$. f is 2-strongly convex and 12-smooth. GD converges for $0 < \alpha < 2/L = \frac{1}{6}$; the standard step is $1/L = \frac{1}{12}$, the optimal step for a quadratic is $2/(L+\mu) = \frac{1}{7}$ with factor $(\kappa-1)/(\kappa+1) = \frac{5}{7}$.

Problem 2: gradient descent

For $f(x,y) = 3(x-1)^2 + y^2$, perform three steps of gradient descent from $x_0 = (5,\ -8)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (6(x-1),\ 2y)$, $H = \begin{pmatrix} 6 & 0 \\ 0 & 2 \end{pmatrix}$, eigenvalues $\mu = 2$, $L = 6$. Minimizer $x^\star = (1,\ 0)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{4} < 2/L = \frac{1}{3}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{4}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(5,\ -8)$	112	$(24,\ -16)$
1	$(-1,\ -4)$	28	$(-12,\ -8)$
2	$(2,\ -2)$	7	$(6,\ -4)$
3	$\left(\frac{1}{2},\ -1\right)$	$\frac{7}{4}$	—

Why it behaves this way. Each step multiplies the deviation from the minimizer in each coordinate by $1 - \alpha\lambda$: in x by $1 - \frac{1}{4}\cdot 6 = -\frac{1}{2}$, in y by $1 - \frac{1}{4}\cdot 2 = \frac{1}{2}$. A negative factor $-\frac{1}{2}$ causes overshooting past the minimizer (zig-zag), but $|-\frac{1}{2}| < 1$, so the oscillations die out. The overall speed is set by the largest $|1-\alpha\lambda| = \frac{1}{2}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

Ticket 13

Definitions

1. **Stationary point and saddle point.** A stationary point satisfies $\nabla f(x) = 0$. A saddle point is a stationary point that is neither a local minimum nor a local maximum (there are directions of both increase and decrease; e.g. an indefinite Hessian, as for $x^2 - y^2$ at the origin).
2. **Positive definite and positive semidefinite matrix.** Symmetric H: $H \succeq 0$ if $x^\top H x \geq 0$ for all x; $H \succ 0$ if $x^\top H x > 0$ for all $x \neq 0$. Equivalently, all eigenvalues are ≥ 0 (respectively > 0). For 2×2, Sylvester's criterion: $H \succ 0 \iff h_{11} > 0,\ \det H > 0$.
3. **Feasible direction.** A vector d is feasible at $x \in \mathcal{X}$ if $x + td \in \mathcal{X}$ for all sufficiently small $t \geq 0$. Necessary condition for a minimum: $\langle \nabla f(x^\star), d \rangle \geq 0$ for all feasible d.
4. **Descent direction.** d is a strict descent direction at x if $\langle \nabla f(x), d \rangle < 0$. Then $f(x + \alpha d) < f(x)$ for all sufficiently small $\alpha > 0$. The negative gradient $-\nabla f(x)$ is the direction of steepest descent in the Euclidean norm.
5. **Gradient descent with a constant step.** $x_{k+1} = x_k - \alpha \nabla f(x_k)$, $\alpha > 0$. For L-smooth f, any step $0 < \alpha < 2/L$ guarantees decrease; the standard choice is $\alpha = 1/L$.
6. **Karush–Kuhn–Tucker (KKT) conditions.** Stationarity $\nabla_x \mathcal{L}(x^\star, \lambda^\star, \nu^\star) = 0$; primal feasibility $g_i(x^\star) \leq 0$, $h_j(x^\star) = 0$; dual feasibility $\lambda_i^\star \geq 0$; complementary slackness $\lambda_i^\star g_i(x^\star) = 0$. For convex f, g_i and affine h_j they are sufficient for global optimality.

Theoretical results

1. First-order necessary condition: at an unconstrained local minimizer $\nabla f(x^\star) = 0$.

Proof. Fix a direction d and let $\phi(t) = f(x^\star + td)$. Since $x^\star$ is a local minimizer of f, $t = 0$ is a local minimizer of the one-variable function ϕ, so $\phi'(0) = 0$. By the chain rule, $\phi'(0) = \langle \nabla f(x^\star), d \rangle = 0$ for every d. Taking $d = \nabla f(x^\star)$ gives $\|\nabla f(x^\star)\|^2 = 0$, i.e. $\nabla f(x^\star) = 0$. □

2. Stability of GD on a quadratic $\frac{1}{2} x^\top Q x - b^\top x$, $Q \succ 0$: convergence from every x_0 $\iff$ $0 < \alpha < 2/\lambda_{\max}(Q)$.

Proof. For the error $e_k = x_k - x^\star$ (where $Q x^\star = b$): $e_{k+1} = e_k - \alpha Q e_k = (I - \alpha Q) e_k$. In the eigenbasis $Q = U \Lambda U^\top$ the coordinates $z = U^\top e$ evolve independently: $z_{k+1,i} = (1 - \alpha \lambda_i) z_{k,i}$. Each coordinate tends to zero iff $|1 - \alpha \lambda_i| < 1$, i.e. $0 < \alpha < 2/\lambda_i$. The tightest restriction comes from $\lambda_{\max}$. □

Problem 1: type of a stationary point

Find the stationary points of $f(x, y) = x^2 + 3y^2 - 2x + 18y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x} = 2x - 2 = 0,\quad \dfrac{\partial f}{\partial y} = 6y + 18 = 0$. The system is linear with determinant $12 \neq 0$, so the unique solution is $x^\star = (1,\ -3)$.

2. Hessian. $H = \begin{pmatrix} 2 & 0 \\ 0 & 6 \end{pmatrix}$ is diagonal with positive entries, hence $H \succ 0$.

Via eigenvalues: $\lambda^2 - 8\lambda + 12 = 0,\ \lambda_1 = 2,\ \lambda_2 = 6$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H (x - x^\star) > 0$ for $x \neq x^\star$, so the minimum is global and unique. $f(x^\star) = -28$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 2$, $L = \lambda_{\max}(H) = 6$, $\kappa = L/\mu = 3$. f is 2-strongly convex and 6-smooth. GD converges for $0 < \alpha < 2/L = \frac{1}{3}$; the standard step is $1/L = \frac{1}{6}$, the optimal step for a quadratic is $2/(L+\mu) = \frac{1}{4}$ with factor $(\kappa-1)/(\kappa+1) = \frac{1}{2}$.

Problem 2: gradient descent

For $f(x,y) = x^2 + xy + y^2$, perform three steps of gradient descent from $x_0 = (3,\ 0)$ with step $\alpha = \frac{1}{3}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (2x + y,\ x + 2y)$, $H = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, eigenvalues $\mu = 1$, $L = 3$. Minimizer $x^\star = (0,\ 0)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{3} < 2/L = \frac{2}{3}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{3}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(3,\ 0)$	9	$(6,\ 3)$
1	$(1,\ -1)$	1	$(1,\ -1)$
2	$\left(\frac{2}{3},\ -\frac{2}{3}\right)$	$\frac{4}{9}$	$\left(\frac{2}{3},\ -\frac{2}{3}\right)$
3	$\left(\frac{4}{9},\ -\frac{4}{9}\right)$	$\frac{16}{81}$	—

Why it behaves this way. Eigendirections of H: $(1,1)$ with $\lambda = 3$ and $(1,-1)$ with $\lambda = 1$. Each step multiplies the error components along them by $1 - \frac{1}{3}\cdot 3 = 0$ and $1 - \frac{1}{3}\cdot 1 = \frac{2}{3}$ respectively. A factor of 0 means this component is eliminated in a single step (step $\alpha = 1/\lambda$). The overall speed is set by the largest $|1 - \alpha\lambda| = \frac{2}{3}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{2}{3}\right)^2 = \frac{4}{9}$.

Ticket 14

Definitions

1. **Optimization problem: decision variable, objective, feasible set.** A problem $\min_{x\in\mathcal{X}} f(x)$, where x is the decision variable (e.g. model weights), f is the objective (error, cost, risk), and $\mathcal{X}$ is the feasible set (constraints; without constraints $\mathcal{X}=\mathbb{R}^d$). Maximizing f is equivalent to minimizing $-f$.
2. **Strict local minimum.** Like a local minimum, but with strict inequality: $f(x^\star) < f(x)$ for all $x \in \mathcal{X}\cap B_r(x^\star)$, $x \neq x^\star$.
3. **Directional derivative.** $D_d f(x) = \lim_{t\downarrow 0} \frac{f(x+td)-f(x)}{t}$. For differentiable f: $D_d f(x) = \langle \nabla f(x), d\rangle$.
4. **Convex function.** f on a convex set C is convex if $f((1-t)x+ty) \le (1-t)f(x)+tf(y)$ for all $x,y\in C$, $t\in[0,1]$. For differentiable f this is equivalent to $f(y)\ge f(x)+\langle\nabla f(x), y-x\rangle$, and for C^2 functions to $\nabla^2 f \succeq 0$.
5. **Polyak–Lojasiewicz condition.** f satisfies PL with $\mu>0$ if $\frac{1}{2}\|\nabla f(x)\|^2 \ge \mu\,(f(x)-f^\star)$ for all x. It requires neither convexity nor a unique minimizer (example: $f(x,y)=x^2$).
6. **Active constraint.** An inequality $g_i(x)\le 0$ is active at a feasible x if $g_i(x)=0$. Inactive constraints impose no local restriction; by complementary slackness an inactive constraint has $\lambda_i^\star = 0$.

Theoretical results

1. For $f(x)=\frac{1}{2}x^\top Ax - b^\top x + c$, $A=A^\top \succ 0$: $\nabla f = Ax-b$, $\nabla^2 f = A$, and $x^\star = A^{-1}b$ is the unique global minimizer.
Proof. $df = \frac{1}{2}(dx^\top Ax + x^\top A\,dx) - b^\top dx = \langle Ax-b, dx\rangle$ (using symmetry of A), so $\nabla f = Ax-b$; differentiating once more gives $\nabla^2 f = A$. Stationarity: $Ax=b$, $x^\star = A^{-1}b$. Substituting $b=Ax^\star$ yields the identity $f(x)-f(x^\star) = \frac{1}{2}(x-x^\star)^\top A(x-x^\star)$, which is >0 for $x\neq x^\star$ since $A\succ 0$. □

2. Newton's method reaches the minimizer of a quadratic with $Q\succ 0$ in one step.
Proof. For $f(x)=\frac{1}{2}x^\top Qx - b^\top x + c$: $\nabla f(x) = Qx-b$, $\nabla^2 f = Q$. The Newton equation $Qp = -(Qx-b)$ gives $p = -x + Q^{-1}b = x^\star - x$. Hence $x^+ = x+p = x^\star$ from any starting point. □

Problem 1: type of a stationary point

Find the stationary points of $f(x,y)=5x^2-4xy+2y^2-16x+4y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x} = 10x-4y-16=0, \quad \dfrac{\partial f}{\partial y} = -4x+4y+4=0$. The system is linear with determinant $24\neq 0$, so the unique solution is $x^\star = (2,\ 1)$.

2. Hessian. $H = \begin{pmatrix} 10 & -4 \\ -4 & 4 \end{pmatrix}$ (constant). Sylvester's criterion: $h_{11} = 10 > 0$, $\det H = 10\cdot 4 - 4^2 = 24 > 0$, hence $H\succ 0$.

Via eigenvalues: $\lambda^2 - 14\lambda + 24 = 0$, $\lambda_1 = 2,\ \lambda_2 = 12$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H(x - x^\star) > 0$ for $x \neq x^\star$, so the minimum is global and unique. $f(x^\star) = -14$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 2$, $L = \lambda_{\max}(H) = 12$, $\kappa = L/\mu = 6$. f is 2-strongly convex and 12-smooth. GD converges for $0 < \alpha < 2/L = \frac{1}{6}$; the standard step is $1/L = \frac{1}{12}$, the optimal step for a quadratic is $2/(L + \mu) = \frac{1}{7}$ with factor $(\kappa - 1)/(\kappa + 1) = \frac{5}{7}$.

Problem 2: gradient descent

For $f(x, y) = x^2 + 4y^2$, perform three steps of gradient descent from $x_0 = (16,\ 1)$ with step $\alpha = \frac{1}{8}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (2x,\ 8y)$, $H = \begin{pmatrix} 2 & 0 \\ 0 & 8 \end{pmatrix}$, eigenvalues $\mu = 2$, $L = 8$. Minimizer $x^\star = (0,\ 0)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{8} < 2/L = \frac{1}{4}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{8}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(16,\ 1)$	260	$(32,\ 8)$
1	$(12,\ 0)$	144	$(24,\ 0)$
2	$(9,\ 0)$	81	$(18,\ 0)$
3	$\left(\frac{27}{4},\ 0\right)$	$\frac{729}{16}$	—

Why it behaves this way. Each step multiplies the deviation from the minimizer in each coordinate by $1 - \alpha\lambda$: in x by $1 - \frac{1}{8} \cdot 2 = \frac{3}{4}$, in y by $1 - \frac{1}{8} \cdot 8 = 0$. A factor of 0 means this component is eliminated in a single step (step $\alpha = 1/\lambda$). The overall speed is set by the largest $|1 - \alpha\lambda| = \frac{3}{4}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{3}{4}\right)^2 = \frac{9}{16}$.

Ticket 15

Definitions

1. **Global minimum.** $x^\star \in \mathcal{X}$ is a global minimizer if $f(x^\star) \le f(x)$ for all $x \in \mathcal{X}$. A minimum may fail to exist (e^x on $\mathbb{R}$) and may be non-unique ($(x^2-1)^2$).
2. **Gradient (via the linear approximation).** For differentiable $f : \mathbb{R}^d \to \mathbb{R}$, the gradient $\nabla f(x)$ is the vector giving the linear part of the increment: $f(x+h) = f(x) + \langle \nabla f(x), h\rangle + o(\|h\|)$. Equivalently, $df = \langle \nabla f(x), dx\rangle$.
3. **Convex combination and convex set.** A convex combination is $\sum_i \lambda_i x_i$ with $\lambda_i \ge 0$, $\sum_i \lambda_i = 1$. A set C is convex if $(1-t)x + ty \in C$ for all $x, y \in C$, $t \in [0,1]$ (it contains the segment between any two of its points).
4. **L-smoothness (Lipschitz gradient).** $\|\nabla f(x) - \nabla f(y)\| \le L\|x-y\|$ for all x, y. For $f \in C^2$ it suffices that $\|\nabla^2 f(x)\|_2 \le L$. Consequence (descent lemma): $f(y) \le f(x) + \langle \nabla f(x), y-x\rangle + \frac{L}{2}\|y-x\|^2$. For a quadratic, $L = \lambda_{\max}(Q)$.
5. **Armijo condition (backtracking).** A step α is accepted if $f(x+\alpha d) \le f(x) + c\alpha\langle \nabla f(x), d\rangle$, $c \in (0,1)$ (typically $c = 10^{-4}$). Backtracking: start from α_0 and shrink $\alpha \leftarrow \beta\alpha$, $\beta \in (0,1)$, until the condition holds.
6. **Dual function and dual problem.** $q(\lambda,\nu) = \inf_x \mathcal{L}(x,\lambda,\nu)$ is always concave (an infimum of affine functions). Dual problem: $\max_{\lambda\ge 0,\nu} q(\lambda,\nu)$. Weak duality: $d^\star \le p^\star$.

Theoretical results

1. Strong convexity implies the Polyak–Lojasiewicz condition.
Proof. By μ-strong convexity, $f(y) \ge f(x) + \langle \nabla f(x), y-x\rangle + \frac{\mu}{2}\|y-x\|^2$ for all y. Minimizing the right side over y: the minimum is attained at $y = x - \frac{1}{\mu}\nabla f(x)$ and equals $f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2$. Since the left side at $y = x^\star$ equals $f^\star$, we get $f^\star \ge f(x) - \frac{1}{2\mu}\|\nabla f(x)\|^2$, i.e. $\frac{1}{2}\|\nabla f(x)\|^2 \ge \mu(f(x) - f^\star)$. □

2. Linear convergence of GD under PL: with $\alpha = 1/L$, $f(x_k) - f^\star \le (1-\mu/L)^k (f(x_0) - f^\star)$.
Proof. By L-smoothness (descent lemma) with $\alpha = 1/L$: $f(x_{k+1}) \le f(x_k) - \frac{1}{2L}\|\nabla f(x_k)\|^2$. By PL, $\|\nabla f(x_k)\|^2 \ge 2\mu(f(x_k) - f^\star)$. Therefore $f(x_{k+1}) \le f(x_k) - \frac{\mu}{L}(f(x_k) - f^\star)$. Subtracting $f^\star$: $f(x_{k+1}) - f^\star \le (1 - \frac{\mu}{L})(f(x_k) - f^\star)$; iterate. Convexity was never used. □

Problem 1: type of a stationary point

Find the stationary points of $f(x,y) = 6x^2 + 4xy + 3y^2 - 16x - 10y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x} = 12x + 4y - 16 = 0, \quad \dfrac{\partial f}{\partial y} = 4x + 6y - 10 = 0$. The system is linear with determinant $56 \ne 0$, so the unique solution is $x^\star = (1,\ 1)$.

2. Hessian. $H = \begin{pmatrix} 12 & 4 \\ 4 & 6 \end{pmatrix}$ (constant). Sylvester's criterion: $h_{11} = 12 > 0$, $\det H = 12 \cdot 6 - 4^2 = 56 > 0$, hence $H \succ 0$.

Via eigenvalues: $\lambda^2 - 18\lambda + 56 = 0$, $\lambda_1 = 4,\ \lambda_2 = 14$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H (x - x^\star) > 0$ for $x \ne x^\star$, so the minimum is global and unique. $f(x^\star) = -13$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 4$, $L = \lambda_{\max}(H) = 14$, $\kappa = L/\mu = \frac{7}{2}$. f is 4-strongly convex and 14-smooth. GD converges for $0 < \alpha < 2/L = \frac{1}{7}$; the standard step is $1/L = \frac{1}{14}$, the optimal step for a quadratic is $2/(L+\mu) = \frac{1}{9}$ with factor $(\kappa-1)/(\kappa+1) = \frac{5}{9}$.

Problem 2: gradient descent

For $f(x,y) = x^2 + xy + y^2$, perform three steps of gradient descent from $x_0 = (0,\ 8)$ with step $\alpha = \frac{1}{2}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (2x + y,\ x + 2y)$, $H = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, eigenvalues $\mu = 1$, $L = 3$. Minimizer $x^\star = (0,\ 0)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{2} < 2/L = \frac{2}{3}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{2}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(0,\ 8)$	64	$(8,\ 16)$
1	$(-4,\ 0)$	16	$(-8,\ -4)$
2	$(0,\ 2)$	4	$(2,\ 4)$
3	$(-1,\ 0)$	1	—

Why it behaves this way. Eigendirections of H: $(1,1)$ with $\lambda = 3$ and $(1,-1)$ with $\lambda = 1$. Each step multiplies the error components along them by $1 - \frac{1}{2}\cdot 3 = -\frac{1}{2}$ and $1 - \frac{1}{2}\cdot 1 = \frac{1}{2}$ respectively. A negative factor $-\frac{1}{2}$ causes overshooting past the minimizer (zig-zag), but $|-\frac{1}{2}| < 1$, so the oscillations die out. The overall speed is set by the largest $|1 - \alpha\lambda| = \frac{1}{2}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

Ticket 16

Definitions

1. **Local minimum.** $x^\star \in \mathcal{X}$ is a local minimizer if there exists $r > 0$ such that $f(x^\star) \le f(x)$ for all $x \in \mathcal{X} \cap B_r(x^\star)$.
2. **Hessian.** The matrix of second partial derivatives $\nabla^2 f(x) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right)_{i,j}$. It is symmetric when the mixed derivatives are continuous (Clairaut). The quantity $d^\top \nabla^2 f(x) d$ is the curvature along d.
3. **Strictly convex and μ-strongly convex function.** Strictly convex: the convexity inequality is strict for $x \neq y$, $t \in (0,1)$. μ-strongly convex ($\mu > 0$): $f((1-t)x+ty) \le (1-t)f(x)+tf(y)-\frac{\mu}{2}t(1-t)\|x-y\|^2$; for differentiable f: $f(y) \ge f(x)+\langle \nabla f(x), y-x\rangle+\frac{\mu}{2}\|y-x\|^2$; for C^2 it suffices that $\nabla^2 f \succeq \mu I$. Example: x^4 is strictly but not strongly convex.
4. **Condition number.** For an L-smooth μ-strongly convex function, $\kappa = L/\mu \ge 1$; for a quadratic, $\kappa = \lambda_{\max}(Q)/\lambda_{\min}(Q)$. The larger κ, the more elongated the level sets and the slower gradient descent (contraction factor $1 - 1/\kappa$ for $\alpha = 1/L$).
5. **Lagrangian.** For $\min f(x)$ subject to $g_i(x) \le 0$, $h_j(x) = 0$: $\mathcal{L}(x, \lambda, \nu) = f(x)+\sum_i \lambda_i g_i(x)+\sum_j \nu_j h_j(x)$, where $\lambda_i \ge 0$ and ν_j are unrestricted in sign.
6. **Newton's method (Newton direction).** Minimizes the quadratic model $m(p) = f(x) + g^\top p + \frac{1}{2}p^\top H p$, $g = \nabla f(x)$, $H = \nabla^2 f(x)$: the direction p_N solves $H p_N = -g$, and $x^+ = x + p_N$ (in practice with damping). Near a nondegenerate minimizer it converges quadratically.

Theoretical results

1. A symmetric $H \succeq 0$ if and only if all its eigenvalues are nonnegative.
Proof. By the spectral theorem $H = Q\Lambda Q^\top$ with orthogonal Q. For $z = Q^\top x$: $x^\top H x = z^\top \Lambda z = \sum_i \lambda_i z_i^2$. If all $\lambda_i \ge 0$, the sum is nonnegative. Conversely, for an eigenvector q_i: $q_i^\top H q_i = \lambda_i \|q_i\|^2 \ge 0$, so $\lambda_i \ge 0$. The case $H \succ 0$ is analogous. □

2. Guaranteed decrease of a GD step for L-smooth f: $f(x^+) \le f(x)-\alpha(1-\frac{L\alpha}{2})\|\nabla f(x)\|^2$.
Proof. Descent lemma (may be used without proof): $f(y) \le f(x) + \langle \nabla f(x), y - x\rangle + \frac{L}{2}\|y - x\|^2$. Substitute $y = x^+ = x - \alpha \nabla f(x)$: $f(x^+) \le f(x) - \alpha\|\nabla f(x)\|^2 + \frac{L\alpha^2}{2}\|\nabla f(x)\|^2 = f(x) - \alpha\left(1 - \frac{L\alpha}{2}\right)\|\nabla f(x)\|^2$. Hence for $0 < \alpha < 2/L$ there is strict decrease away from stationary points, and for $\alpha = 1/L$: $f(x^+) \le f(x) - \frac{1}{2L}\|\nabla f(x)\|^2$. □

Problem 1: type of a stationary point

Find the stationary points of $f(x,y) = x^2-xy+y^2+x-2y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x} = 2x - y + 1 = 0, \quad \dfrac{\partial f}{\partial y} = -x + 2y - 2 = 0.$ The system is linear with determinant $3 \neq 0$, so the unique solution is $x^\star = (0,\ 1)$.

2. Hessian. $H = \begin{pmatrix} 2 & -1 \\ -1 & 2 \end{pmatrix}$ (constant). Sylvester's criterion: $h_{11} = 2 > 0$, $\det H = 2 \cdot 2 - 1^2 = 3 > 0$, hence $H \succ 0$.

Via eigenvalues: $\lambda^2 - 4\lambda + 3 = 0$, $\lambda_1 = 1,\ \lambda_2 = 3$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H(x - x^\star) > 0$ for $x \neq x^\star$, so the minimum is global and unique. $f(x^\star) = -1$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 1$, $L = \lambda_{\max}(H) = 3$, $\kappa = L/\mu = 3$. f is 1-strongly convex and 3-smooth. GD converges for $0 < \alpha < 2/L = \frac{2}{3}$; the standard step is $1/L = \frac{1}{3}$, the optimal step for a quadratic is $2/(L + \mu) = \frac{1}{2}$ with factor $(\kappa - 1)/(\kappa + 1) = \frac{1}{2}$.

Problem 2: gradient descent

For $f(x, y) = (x + 1)^2 + 2y^2$, perform three steps of gradient descent from $x_0 = (7,\ 2)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (2(x + 1),\ 4y)$, $H = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$, eigenvalues $\mu = 2$, $L = 4$. Minimizer $x^\star = (-1,\ 0)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{4} < 2/L = \frac{1}{2}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{4}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(7,\ 2)$	72	$(16,\ 8)$
1	$(3,\ 0)$	16	$(8,\ 0)$
2	$(1,\ 0)$	4	$(4,\ 0)$
3	$(0,\ 0)$	1	—

Why it behaves this way. Each step multiplies the deviation from the minimizer in each coordinate by $1 - \alpha\lambda$: in x by $1 - \frac{1}{4} \cdot 2 = \frac{1}{2}$, in y by $1 - \frac{1}{4} \cdot 4 = 0$. A factor of 0 means this component is eliminated in a single step (step $\alpha = 1/\lambda$). The overall speed is set by the largest $|1 - \alpha\lambda| = \frac{1}{2}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

Ticket 17

Definitions

1. **Gradient (via the linear approximation).** For differentiable $f : \mathbb{R}^d \to \mathbb{R}$, the gradient $\nabla f(x)$ is the vector giving the linear part of the increment: $f(x+h) = f(x) + \langle \nabla f(x), h\rangle + o(\|h\|)$. Equivalently, $df = \langle \nabla f(x), dx\rangle$.
2. **Stationary point and saddle point.** A stationary point satisfies $\nabla f(x) = 0$. A saddle point is a stationary point that is neither a local minimum nor a local maximum (there are directions of both increase and decrease; e.g. an indefinite Hessian, as for $x^2 - y^2$ at the origin).
3. **Positive definite and positive semidefinite matrix.** Symmetric H: $H \succeq 0$ if $x^\top H x \geq 0$ for all x; $H \succ 0$ if $x^\top H x > 0$ for all $x \neq 0$. Equivalently, all eigenvalues are ≥ 0 (respectively > 0). For 2×2, Sylvester's criterion: $H \succ 0 \iff h_{11} > 0,\ \det H > 0$.
4. **L-smoothness (Lipschitz gradient).** $\|\nabla f(x) - \nabla f(y)\| \leq L\|x - y\|$ for all x, y. For $f \in C^2$ it suffices that $\|\nabla^2 f(x)\|_2 \leq L$. Consequence (descent lemma): $f(y) \leq f(x) + \langle \nabla f(x), y - x\rangle + \frac{L}{2}\|y - x\|^2$. For a quadratic, $L = \lambda_{\max}(Q)$.
5. **Lagrangian.** For $\min f(x)$ subject to $g_i(x) \leq 0$, $h_j(x) = 0$: $\mathcal{L}(x, \lambda, \nu) = f(x) + \sum_i \lambda_i g_i(x) + \sum_j \nu_j h_j(x)$, where $\lambda_i \geq 0$ and ν_j are unrestricted in sign.
6. **Newton's method (Newton direction).** Minimizes the quadratic model $m(p) = f(x) + g^\top p + \frac{1}{2} p^\top H p$, $g = \nabla f(x)$, $H = \nabla^2 f(x)$: the direction p_N solves $H p_N = -g$, and $x^+ = x + p_N$ (in practice with damping). Near a nondegenerate minimizer it converges quadratically.

Theoretical results

1. Weak duality: $q(\lambda, \nu) \leq f(x)$ for every feasible x and $\lambda \geq 0$.
Proof. By definition of the infimum, $q(\lambda, \nu) = \inf_z \mathcal{L}(z, \lambda, \nu) \leq \mathcal{L}(x, \lambda, \nu)$. For feasible x and $\lambda \geq 0$: $\sum_i \lambda_i g_i(x) \leq 0$ and $\sum_j \nu_j h_j(x) = 0$, so $\mathcal{L}(x, \lambda, \nu) \leq f(x)$. Hence $q(\lambda, \nu) \leq f(x)$; taking the supremum on the left and the infimum on the right gives $d^\star \leq p^\star$. □

2. The negative gradient is the direction of steepest local decrease (among $\|d\|_2 = 1$).
Proof. The rate of change along a unit d is $D_d f(x) = \langle \nabla f(x), d\rangle$. By Cauchy–Schwarz, $\langle \nabla f(x), d\rangle \geq -\|\nabla f(x)\|\|d\| = -\|\nabla f(x)\|$, with equality for $d = -\nabla f(x)/\|\nabla f(x)\|$. So this direction gives the smallest (most negative) derivative. □

Problem 1: type of a stationary point

Find the stationary points of $f(x, y) = 2x^2 + 2xy + 2y^2 - 8x - 4y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x} = 4x + 2y - 8 = 0, \quad \dfrac{\partial f}{\partial y} = 2x + 4y - 4 = 0$. The system is linear with determinant $12 \neq 0$, so the unique solution is $x^\star = (2,\ 0)$.

2. Hessian. $H = \begin{pmatrix} 4 & 2 \\ 2 & 4 \end{pmatrix}$ (constant). Sylvester's criterion: $h_{11} = 4 > 0$, $\det H = 4 \cdot 4 - 2^2 = 12 > 0$, hence $H \succ 0$.

Via eigenvalues: $\lambda^2 - 8\lambda + 12 = 0,\ \lambda_1 = 2,\ \lambda_2 = 6$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H(x - x^\star) > 0$ for $x \neq x^\star$, so the minimum is global and unique. $f(x^\star) = -8$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 2$, $L = \lambda_{\max}(H) = 6$, $\kappa = L/\mu = 3$. f is 2-strongly convex and 6-smooth. GD converges for $0 < \alpha < 2/L = \frac{1}{3}$; the standard step is $1/L = \frac{1}{6}$, the optimal step for a quadratic is $2/(L + \mu) = \frac{1}{4}$ with factor $(\kappa - 1)/(\kappa + 1) = \frac{1}{2}$.

Problem 2: gradient descent

For $f(x, y) = 2x^2 + (y - 2)^2$, perform three steps of gradient descent from $x_0 = (4,\ 6)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (4x,\ 2(y - 2))$, $H = \begin{pmatrix} 4 & 0 \\ 0 & 2 \end{pmatrix}$, eigenvalues $\mu = 2$, $L = 4$. Minimizer $x^\star = (0,\ 2)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{4} < 2/L = \frac{1}{2}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{4}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(4,\ 6)$	48	$(16,\ 8)$
1	$(0,\ 4)$	4	$(0,\ 4)$
2	$(0,\ 3)$	1	$(0,\ 2)$
3	$\left(0,\ \frac{5}{2}\right)$	$\frac{1}{4}$	—

Why it behaves this way. Each step multiplies the deviation from the minimizer in each coordinate by $1 - \alpha\lambda$: in x by $1 - \frac{1}{4} \cdot 4 = 0$, in y by $1 - \frac{1}{4} \cdot 2 = \frac{1}{2}$. A factor of 0 means this component is eliminated in a single step (step $\alpha = 1/\lambda$). The overall speed is set by the largest $|1 - \alpha\lambda| = \frac{1}{2}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

Ticket 18

Definitions

1. **Strict local minimum.** Like a local minimum, but with strict inequality: $f(x^\star) < f(x)$ for all $x \in \mathcal{X} \cap B_r(x^\star)$, $x \neq x^\star$.
2. **Feasible direction.** A vector d is feasible at $x \in \mathcal{X}$ if $x + td \in \mathcal{X}$ for all sufficiently small $t \geq 0$. Necessary condition for a minimum: $\langle \nabla f(x^\star), d\rangle \geq 0$ for all feasible d.
3. **Convex combination and convex set.** A convex combination is $\sum_i \lambda_i x_i$ with $\lambda_i \geq 0$, $\sum_i \lambda_i = 1$. A set C is convex if $(1-t)x + ty \in C$ for all $x, y \in C$, $t \in [0,1]$ (it contains the segment between any two of its points).
4. **Descent direction.** d is a strict descent direction at x if $\langle \nabla f(x), d\rangle < 0$. Then $f(x + \alpha d) < f(x)$ for all sufficiently small $\alpha > 0$. The negative gradient $-\nabla f(x)$ is the direction of steepest descent in the Euclidean norm.
5. **Karush–Kuhn–Tucker (KKT) conditions.** Stationarity $\nabla_x \mathcal{L}(x^\star, \lambda^\star, \nu^\star) = 0$; primal feasibility $g_i(x^\star) \leq 0$, $h_j(x^\star) = 0$; dual feasibility $\lambda_i^\star \geq 0$; complementary slackness $\lambda_i^\star g_i(x^\star) = 0$. For convex f, g_i and affine h_j they are sufficient for global optimality.
6. **Dual function and dual problem.** $q(\lambda, \nu) = \inf_x \mathcal{L}(x, \lambda, \nu)$ is always concave (an infimum of affine functions). Dual problem: $\max_{\lambda \geq 0, \nu} q(\lambda, \nu)$. Weak duality: $d^\star \leq p^\star$.

Theoretical results

1. Second-order necessary condition: at an unconstrained local minimizer $\nabla^2 f(x^\star) \succeq 0$.
Proof. By the first-order condition, $\nabla f(x^\star) = 0$. Taylor expansion along td: $f(x^\star + td) - f(x^\star) = \frac{t^2}{2} d^\top \nabla^2 f(x^\star) d + o(t^2)$. The left side is ≥ 0 for small t. Dividing by $t^2/2$ and letting $t \to 0$ gives $d^\top \nabla^2 f(x^\star) d \geq 0$ for every d, i.e. $\nabla^2 f(x^\star) \succeq 0$. □

2. The intersection of any family of convex sets is convex.
Proof. Let $C = \bigcap_\alpha C_\alpha$, $x, y \in C$, $t \in [0,1]$. Then $x, y \in C_\alpha$ for every α; by convexity of C_α, $(1-t)x + ty \in C_\alpha$ for every α, hence $(1-t)x + ty \in C$. □

Problem 1: type of a stationary point

Find the stationary points of $f(x,y) = 2x^2 - 2xy + 2y^2 + 4x - 8y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x} = 4x - 2y + 4 = 0, \quad \dfrac{\partial f}{\partial y} = -2x + 4y - 8 = 0.$ The system is linear with determinant $12 \neq 0$, so the unique solution is $x^\star = (0,\ 2)$.

2. Hessian. $H = \begin{pmatrix} 4 & -2 \\ -2 & 4 \end{pmatrix}$ (constant). Sylvester's criterion: $h_{11} = 4 > 0$, $\det H = 4 \cdot 4 - 2^2 = 12 > 0$, hence $H \succ 0$.

Via eigenvalues: $\lambda^2 - 8\lambda + 12 = 0$, $\lambda_1 = 2,\ \lambda_2 = 6$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H (x - x^\star) > 0$ for $x \neq x^\star$, so the minimum is global and unique. $f(x^\star) = -8$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2} h^\top H h$; if it is positive in every direction, the function

increases wherever we move (a “bowl”). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 2$, $L = \lambda_{\max}(H) = 6$, $\kappa = L/\mu = 3$. f is 2-strongly convex and 6-smooth. GD converges for $0 < \alpha < 2/L = \frac{1}{3}$; the standard step is $1/L = \frac{1}{6}$, the optimal step for a quadratic is $2/(L+\mu) = \frac{1}{4}$ with factor $(\kappa-1)/(\kappa+1) = \frac{1}{2}$.

Problem 2: gradient descent

For $f(x,y) = x^2 + xy + y^2$, perform three steps of gradient descent from $x_0 = (4,\ 0)$ with step $\alpha = \frac{1}{2}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (2x + y,\ x + 2y)$, $H = \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$, eigenvalues $\mu = 1$, $L = 3$. Minimizer $x^\star = (0,\ 0)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{2} < 2/L = \frac{2}{3}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{2}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(4,\ 0)$	16	$(8,\ 4)$
1	$(0,\ -2)$	4	$(-2,\ -4)$
2	$(1,\ 0)$	1	$(2,\ 1)$
3	$\left(0,\ -\frac{1}{2}\right)$	$\frac{1}{4}$	—

Why it behaves this way. Eigendirections of H: $(1,1)$ with $\lambda = 3$ and $(1,-1)$ with $\lambda = 1$. Each step multiplies the error components along them by $1 - \frac{1}{2}\cdot 3 = -\frac{1}{2}$ and $1 - \frac{1}{2}\cdot 1 = \frac{1}{2}$ respectively. A negative factor $-\frac{1}{2}$ causes overshooting past the minimizer (zig-zag), but $|-\frac{1}{2}| < 1$, so the oscillations die out. The overall speed is set by the largest $|1 - \alpha\lambda| = \frac{1}{2}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

Ticket 19

Definitions

1. **Global minimum.** $x^\star \in \mathcal{X}$ is a global minimizer if $f(x^\star) \le f(x)$ for all $x \in \mathcal{X}$. A minimum may fail to exist (e^x on $\mathbb{R}$) and may be non-unique ($(x^2-1)^2$).
2. **Hessian.** The matrix of second partial derivatives $\nabla^2 f(x) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}\right)_{i,j}$. It is symmetric when the mixed derivatives are continuous (Clairaut). The quantity $d^\top \nabla^2 f(x) d$ is the curvature along d.
3. **Convex function.** f on a convex set C is convex if $f((1-t)x+ty) \le (1-t)f(x)+tf(y)$ for all $x, y \in C$, $t \in [0,1]$. For differentiable f this is equivalent to $f(y) \ge f(x)+\langle \nabla f(x), y-x\rangle$, and for C^2 functions to $\nabla^2 f \succeq 0$.
4. **Strictly convex and μ-strongly convex function.** Strictly convex: the convexity inequality is strict for $x \ne y$, $t \in (0,1)$. μ-strongly convex ($\mu > 0$): $f((1-t)x+ty) \le (1-t)f(x)+tf(y)-\frac{\mu}{2}t(1-t)\|x-y\|^2$; for differentiable f: $f(y) \ge f(x)+\langle \nabla f(x), y-x\rangle+\frac{\mu}{2}\|y-x\|^2$; for C^2 it suffices that $\nabla^2 f \succeq \mu I$. Example: x^4 is strictly but not strongly convex.
5. **Armijo condition (backtracking).** A step α is accepted if $f(x+\alpha d) \le f(x) + c\alpha\langle \nabla f(x), d\rangle$, $c \in (0,1)$ (typically $c = 10^{-4}$). Backtracking: start from α_0 and shrink $\alpha \leftarrow \beta\alpha$, $\beta \in (0,1)$, until the condition holds.
6. **Active constraint.** An inequality $g_i(x) \le 0$ is active at a feasible x if $g_i(x) = 0$. Inactive constraints impose no local restriction; by complementary slackness an inactive constraint has $\lambda_i^\star = 0$.

Theoretical results

1. First-order necessary condition: at an unconstrained local minimizer $\nabla f(x^\star) = 0$.

Proof. Fix a direction d and let $\phi(t) = f(x^\star + td)$. Since $x^\star$ is a local minimizer of f, $t = 0$ is a local minimizer of the one-variable function ϕ, so $\phi'(0) = 0$. By the chain rule, $\phi'(0) = \langle \nabla f(x^\star), d\rangle = 0$ for every d. Taking $d = \nabla f(x^\star)$ gives $\|\nabla f(x^\star)\|^2 = 0$, i.e. $\nabla f(x^\star) = 0$. □

2. The negative gradient is the direction of steepest local decrease (among $\|d\|_2 = 1$).

Proof. The rate of change along a unit d is $D_d f(x) = \langle \nabla f(x), d\rangle$. By Cauchy–Schwarz, $\langle \nabla f(x), d\rangle \ge -\|\nabla f(x)\|\|d\| = -\|\nabla f(x)\|$, with equality for $d = -\nabla f(x)/\|\nabla f(x)\|$. So this direction gives the smallest (most negative) derivative. □

Problem 1: type of a stationary point

Find the stationary points of $f(x,y) = 4x^2 + 2xy + 4y^2 - 20x - 20y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x} = 8x + 2y - 20 = 0, \quad \dfrac{\partial f}{\partial y} = 2x + 8y - 20 = 0.$ The system is linear with determinant $60 \ne 0$, so the unique solution is $x^\star = (2,\ 2)$.

2. Hessian. $H = \begin{pmatrix} 8 & 2 \\ 2 & 8 \end{pmatrix}$ (constant). Sylvester's criterion: $h_{11} = 8 > 0$, $\det H = 8 \cdot 8 - 2^2 = 60 > 0$, hence $H \succ 0$.

Via eigenvalues: $\lambda^2 - 16\lambda + 60 = 0,\ \lambda_1 = 6,\ \lambda_2 = 10$ — both positive.

3. Conclusion. $\nabla f(x^\star) = 0$ and $H \succ 0 \Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x) - f(x^\star) = \frac{1}{2}(x - x^\star)^\top H(x - x^\star) > 0$ for $x \neq x^\star$, so the minimum is global and unique. $f(x^\star) = -40$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 6$, $L = \lambda_{\max}(H) = 10$, $\kappa = L/\mu = \frac{5}{3}$. f is 6-strongly convex and 10-smooth. GD converges for $0 < \alpha < 2/L = \frac{1}{5}$; the standard step is $1/L = \frac{1}{10}$, the optimal step for a quadratic is $2/(L + \mu) = \frac{1}{8}$ with factor $(\kappa - 1)/(\kappa + 1) = \frac{1}{4}$.

Problem 2: gradient descent

For $f(x, y) = x^2 + 3y^2$, perform three steps of gradient descent from $x_0 = (8,\ 4)$ with step $\alpha = \frac{1}{4}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (2x,\ 6y)$, $H = \begin{pmatrix} 2 & 0 \\ 0 & 6 \end{pmatrix}$, eigenvalues $\mu = 2$, $L = 6$. Minimizer $x^\star = (0,\ 0)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{4} < 2/L = \frac{1}{3}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{4}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(8,\ 4)$	112	$(16,\ 24)$
1	$(4,\ -2)$	28	$(8,\ -12)$
2	$(2,\ 1)$	7	$(4,\ 6)$
3	$\left(1,\ -\frac{1}{2}\right)$	$\frac{7}{4}$	—

Why it behaves this way. Each step multiplies the deviation from the minimizer in each coordinate by $1 - \alpha\lambda$: in x by $1 - \frac{1}{4} \cdot 2 = \frac{1}{2}$, in y by $1 - \frac{1}{4} \cdot 6 = -\frac{1}{2}$. A negative factor $-\frac{1}{2}$ causes overshooting past the minimizer (zig-zag), but $|-\frac{1}{2}| < 1$, so the oscillations die out. The overall speed is set by the largest $|1 - \alpha\lambda| = \frac{1}{2}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{1}{2}\right)^2 = \frac{1}{4}$.

Ticket 20

Definitions

1. **Optimization problem: decision variable, objective, feasible set.** A problem $\min_{x\in\mathcal{X}} f(x)$, where x is the decision variable (e.g. model weights), f is the objective (error, cost, risk), and $\mathcal{X}$ is the feasible set (constraints; without constraints $\mathcal{X}=\mathbb{R}^d$). Maximizing f is equivalent to minimizing $-f$.
2. **Local minimum.** $x^\star\in\mathcal{X}$ is a local minimizer if there exists $r>0$ such that $f(x^\star)\le f(x)$ for all $x\in\mathcal{X}\cap B_r(x^\star)$.
3. **Directional derivative.** $D_d f(x)=\lim_{t\downarrow 0}\frac{f(x+td)-f(x)}{t}$. For differentiable f: $D_d f(x)=\langle\nabla f(x),d\rangle$.
4. **Polyak–Lojasiewicz condition.** f satisfies PL with $\mu>0$ if $\frac{1}{2}\|\nabla f(x)\|^2\ge\mu\,(f(x)-f^\star)$ for all x. It requires neither convexity nor a unique minimizer (example: $f(x,y)=x^2$).
5. **Condition number.** For an L-smooth μ-strongly convex function, $\kappa=L/\mu\ge 1$; for a quadratic, $\kappa=\lambda_{\max}(Q)/\lambda_{\min}(Q)$. The larger κ, the more elongated the level sets and the slower gradient descent (contraction factor $1-1/\kappa$ for $\alpha=1/L$).
6. **Gradient descent with a constant step.** $x_{k+1}=x_k-\alpha\nabla f(x_k)$, $\alpha>0$. For L-smooth f, any step $0<\alpha<2/L$ guarantees decrease; the standard choice is $\alpha=1/L$.

Theoretical results

1. For $f(x)=\frac{1}{2}x^\top Ax-b^\top x+c$, $A=A^\top\succ 0$: $\nabla f=Ax-b$, $\nabla^2 f=A$, and $x^\star=A^{-1}b$ is the unique global minimizer.
Proof. $df=\frac{1}{2}(dx^\top Ax+x^\top A\,dx)-b^\top dx=\langle Ax-b,dx\rangle$ (using symmetry of A), so $\nabla f=Ax-b$; differentiating once more gives $\nabla^2 f=A$. Stationarity: $Ax=b$, $x^\star=A^{-1}b$. Substituting $b=Ax^\star$ yields the identity $f(x)-f(x^\star)=\frac{1}{2}(x-x^\star)^\top A(x-x^\star)$, which is >0 for $x\neq x^\star$ since $A\succ 0$. □

2. Guaranteed decrease of a GD step for L-smooth f: $f(x^+)\le f(x)-\alpha(1-\frac{L\alpha}{2})\|\nabla f(x)\|^2$.
Proof. Descent lemma (may be used without proof): $f(y)\le f(x)+\langle\nabla f(x),y-x\rangle+\frac{L}{2}\|y-x\|^2$. Substitute $y=x^+=x-\alpha\nabla f(x)$: $f(x^+)\le f(x)-\alpha\|\nabla f(x)\|^2+\frac{L\alpha^2}{2}\|\nabla f(x)\|^2=f(x)-\alpha\left(1-\frac{L\alpha}{2}\right)\|\nabla f(x)\|^2$. Hence for $0<\alpha<2/L$ there is strict decrease away from stationary points, and for $\alpha=1/L$: $f(x^+)\le f(x)-\frac{1}{2L}\|\nabla f(x)\|^2$. □

Problem 1: type of a stationary point

Find the stationary points of $f(x,y)=2x^2+xy+2y^2+3x-3y$ and determine their type. Explain why your conclusion holds. Follow-up: what are μ and L for this function?

1. Stationary points. $\dfrac{\partial f}{\partial x}=4x+y+3=0,\quad \dfrac{\partial f}{\partial y}=x+4y-3=0.$ The system is linear with determinant $15\neq 0$, so the unique solution is $x^\star=(-1,\ 1)$.

2. Hessian. $H=\begin{pmatrix}4 & 1\\ 1 & 4\end{pmatrix}$ (constant). Sylvester's criterion: $h_{11}=4>0$, $\det H=4\cdot 4-1^2=15>0$, hence $H\succ 0$.

Via eigenvalues: $\lambda^2-8\lambda+15=0$, $\lambda_1=3,\ \lambda_2=5$ — both positive.

3. Conclusion. $\nabla f(x^\star)=0$ and $H\succ 0\Rightarrow x^\star$ is a strict local minimizer; since f is quadratic, $f(x)-f(x^\star)=\frac{1}{2}(x-x^\star)^\top H(x-x^\star)>0$ for $x\neq x^\star$, so the minimum is global and unique.

$f(x^\star) = -3$.

Intuition: at a stationary point the linear Taylor term vanishes, so the behaviour is governed by the quadratic form $\frac{1}{2}h^\top Hh$; if it is positive in every direction, the function increases wherever we move (a "bowl"). A negative eigenvalue would give a direction of decrease (a saddle or a maximum).

Follow-up (μ, L). $\mu = \lambda_{\min}(H) = 3$, $L = \lambda_{\max}(H) = 5$, $\kappa = L/\mu = \frac{5}{3}$. f is 3-strongly convex and 5-smooth. GD converges for $0 < \alpha < 2/L = \frac{2}{5}$; the standard step is $1/L = \frac{1}{5}$, the optimal step for a quadratic is $2/(L+\mu) = \frac{1}{4}$ with factor $(\kappa-1)/(\kappa+1) = \frac{1}{4}$.

Problem 2: gradient descent

For $f(x,y) = x^2 + 2y^2$, perform three steps of gradient descent from $x_0 = (9,\ 9)$ with step $\alpha = \frac{1}{3}$: write down the gradients, the points x_1, x_2, x_3 and the values of f. Find L and check that the step is safe. Explain how the coordinates behave.

Gradient and Hessian. $\nabla f = (2x,\ 4y)$, $H = \begin{pmatrix} 2 & 0 \\ 0 & 4 \end{pmatrix}$, eigenvalues $\mu = 2$, $L = 4$. Minimizer $x^\star = (0,\ 0)$, $f^\star = 0$.

Step check. $\alpha = \frac{1}{3} < 2/L = \frac{1}{2}$ — the step is safe and the method converges.

Iterations $x_{k+1} = x_k - \frac{1}{3}\nabla f(x_k)$:

k	x_k	$f(x_k)$	$\nabla f(x_k)$
0	$(9,\ 9)$	243	$(18,\ 36)$
1	$(3,\ -3)$	27	$(6,\ -12)$
2	$(1,\ 1)$	3	$(2,\ 4)$
3	$\left(\frac{1}{3},\ -\frac{1}{3}\right)$	$\frac{1}{3}$	—

Why it behaves this way. Each step multiplies the deviation from the minimizer in each coordinate by $1 - \alpha\lambda$: in x by $1 - \frac{1}{3}\cdot 2 = \frac{1}{3}$, in y by $1 - \frac{1}{3}\cdot 4 = -\frac{1}{3}$. A negative factor $-\frac{1}{3}$ causes overshooting past the minimizer (zig-zag), but $|-\frac{1}{3}| < 1$, so the oscillations die out. The overall speed is set by the largest $|1-\alpha\lambda| = \frac{1}{3}$; each step multiplies $f(x_k) - f^\star$ by at most $\left(\frac{1}{3}\right)^2 = \frac{1}{9}$.